

Feldman–Hájek, but Non-Gaussian: Geometry of the Weighted Fourier Jet

Anonymous Author(s)

Abstract

Feldman–Hájek theory gives an exact measure-class classification for Gaussian measures, but what replaces it for a non-Gaussian product law under arbitrary linear mixing? Kakutani’s criterion decides equivalence of infinite product measures, but it breaks down under bounded invertible linear mixing. We give one consolidated, complete answer and it is rigid.

Contents

1	Introduction	2
1.1	The finite-variance classification	2
1.2	Relation to earlier criteria	3
1.3	Two boundary regimes	4
1.4	Proof strategy and organization	4
2	Dimension-Free Hellinger Geometry	5
2.1	A product of centered complex exponentials	5
2.2	Gaussian characterization by a global linear ODE	5
2.3	The determinant representation and finite detection	6
2.4	Dimension-free variance and affine normalization	7
2.5	The matrix Fisher operator	7
2.6	Hellinger paths, polar decomposition, and symmetric perturbations	8
2.7	Finite real Fourier sums with dual moments	9
2.8	Uniform coercivity modulo the exact linear stabilizer	13
3	The Finite-Variance Classification	13
3.1	Measurable row-series realizations on countable operator orbits	14
3.2	Rational translation invariance and the zero–one dichotomy	15
3.3	Finite-marginal likelihood ratios and projected Hellinger energy	16
3.4	Sufficiency of Hilbert–Schmidt perturbations	17
3.4.1	The orthogonal case requires only location Fisher information	18
3.5	Necessity of Hilbert–Schmidt perturbations	19
3.5.1	Characteristic-function rigidity and bounded invertible linear mixing	19
3.6	Exact identification of the linear stabilizer	25
3.7	Two-sided uniform integrability and completion of the proof	25
3.8	Globalization of finite-dimensional coercivity	26

4	Symmetric Stable Products	27
4.1	Anisotropic stable Hellinger coercivity and sharpness	35
5	Density Zeros and Support Rigidity	36
5.1	The metric input when the density has zeros	37
5.2	The linear equivalence group for arbitrary zero sets of the density	38
5.3	Boundary behavior of the density and summability criteria	39
5.4	Complete linear classification for bounded and half-line laws	40
5.5	Why the zero-set row condition is indispensable	41
5.6	Exact block criteria without equivalence of the factors	42
5.7	An exact finite-marginal equivalence criterion	44
5.8	What the Finite-Variance Hypotheses Control	45

1 Introduction

Let ρ be a non-Gaussian probability law and $\mu = \rho^{\otimes \mathbb{N}}$. We ask which bounded invertible operators on ℓ^2 preserve the measure class of μ when their rows act on the coordinate sequence. Kakutani's theorem answers the corresponding question for coordinatewise transformations, but general linear mixing destroys the independence on which its product criterion relies. Fixed finite output marginals do not resolve the problem: they may all have positive densities even when the full image law is singular.

For finite-variance coordinate laws satisfying the regularity assumptions below, the answer has the same operator-ideal form as the Feldman–Hájek criterion, but with a discrete non-Gaussian stabilizer. The image law is equivalent to μ precisely when the mixing operator is a Hilbert–Schmidt perturbation of an allowed signed coordinate permutation. We also show that this operator condition is detected exactly by the Hellinger energies of the actual finite output marginals.

This classification has two sharp boundaries. Infinite variance changes the metric cost: for symmetric stable products, off-diagonal columns accumulate with exponent α , not quadratically. Zeros of a finite-variance density create a different obstruction: Hilbert–Schmidt proximity still controls accumulated Hellinger loss, but equivalence additionally requires the rows of the operator and its inverse to avoid the zero set. Thus tail geometry and support geometry enter the classification in distinct ways.

1.1 The finite-variance classification

Suppose $\rho(dx) = f(x) dx$, put $g = \sqrt{f}$, and assume

$$\begin{aligned} f > 0 \quad \text{Lebesgue-almost everywhere,} \quad g &\in W^{1,2}(\mathbb{R}), \quad xg' \in L^2(\mathbb{R}), \\ \mathbb{E}X &= 0, \quad \mathbb{E}X^2 = 1, \quad f \neq (2\pi)^{-1/2}e^{-x^2/2}. \end{aligned} \tag{1}$$

Derivatives are weak derivatives. These hypotheses impose finite location and scale Fisher information but no moment above order two; in particular, cusps are allowed.

Put $H = \ell^2(\mathbb{N}; \mathbb{R})$, let $(e_i)_{i \geq 1}$ be its coordinate basis, and write $\mathcal{S}_2(H)$ for the Hilbert–Schmidt ideal, with the standard operator-ideal conventions of [35]. Define

$$\begin{aligned} \mathcal{P} = \{Q : (Qx)_i &= \varepsilon_i x_{\pi(i)}, \quad \pi \text{ a permutation of } \mathbb{N}, \\ \varepsilon_i &\in \{-1, 1\}, \quad \text{Law}(\varepsilon_i X) = \rho \text{ for every } i\}. \end{aligned} \tag{2}$$

Thus $\mathcal{P}$ is the full signed-permutation group when ρ is symmetric and the ordinary permutation group otherwise. For $T \in \mathcal{B}(H)$, let Φ_T be the measurable row-series realization constructed in lemma 3.1, and set $\nu_T = (\Phi_T)_\# \mu$. A sample from μ is not assumed to lie in H .

For probability measures λ, ν , write

$$A(\lambda, \nu) = \int \sqrt{\frac{d\lambda}{dm} \frac{d\nu}{dm}} dm, \quad h(\lambda, \nu)^2 = 2 - 2A(\lambda, \nu), \quad (3)$$

where m dominates both measures. Let $\mu_n = \rho^{\otimes n}$, let π_n extract the first n coordinates, and define

$$E_n(T) = -2 \log A(\mu_n, (\pi_n)_\# \nu_T). \quad (4)$$

These energies use the true output rows of T , rather than principal matrix corners.

Theorem 1.1 (Bounded invertible linear mixing). *Under (1), every $T \in \text{GL}(H)$ satisfies*

$$\boxed{\begin{aligned} \nu_T \sim \mu &\iff \exists Q \in \mathcal{P} : TQ^{-1} - I \in \mathcal{S}_2(H) \\ &\iff \sup_n E_n(T) < \infty. \end{aligned}} \quad (5)$$

If these conditions fail, then $\nu_T \perp \mu$ and $E_n(T) \uparrow \infty$. Moreover, $\nu_T = \mu$ if and only if $T \in \mathcal{P}$.

The theorem identifies both the exact symmetry group and the full measure-class neighborhood around it. The projected-energy equivalence is not a finite-dimensional surrogate: bounded energy first detects a nonzero absolutely continuous component, and an orbit dichotomy upgrades that overlap to equivalence.

Two consequences separate the algebraic and analytic parts of the result.

Corollary 1.2 (Moment-only stabilizer). *If ρ is non-Gaussian, centered, and has variance one, then the exact linear stabilizer of $\rho^{\otimes \mathbb{N}}$ is $\mathcal{P}$. No density or Fisher information assumption is needed.*

Corollary 1.3 (Orthogonal transformations). *If $f > 0$ almost everywhere, $\sqrt{f} \in W^{1,2}(\mathbb{R})$, and f is non-Gaussian, centered, and has variance one, then every orthogonal U satisfies*

$$\nu_U \sim \mu \iff U - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}.$$

Thus scale Fisher information is not needed for orthogonal mixing.

1.2 Relation to earlier criteria

Kakutani's theorem reduces equivalence of products with equivalent factors to summability of one-coordinate Hellinger losses [17]. For translations, Shepp relates this summability to the regularity of the square-root density [32]. Euclidean and Lie-group perturbations likewise retain the product structure and identify Fisher information as the local metric [24, 37]. These criteria supply the translation and coordinatewise inputs to our proof; general output rows require estimates that also detect the dependence created by mixing.

In the Gaussian setting, the Cameron–Martin theorem describes admissible translations [7], while the Feldman–Hájek theory and Gaussian likelihood formulas describe equivalence under linear transformations [6, 10, 12, 33]. Ramer treats nonlinear transformations of Gaussian

measures [28]. For the standard Gaussian product, the full orthogonal group consists of exact symmetries. The non-Gaussian result identifies a different stabilizer and measures the distance to that stabilizer, rather than only the covariance defect.

Linear admissibility beyond Gaussian measures was studied by Shimomura [34]. His Theorem 3.4 shows that $I - S^2$ is compact for a self-adjoint admissible operator S , under ℓ^2 -continuity, ℓ^2 -translation quasi-invariance, and ℓ^2 -ergodicity assumptions; his examples also show that those assumptions alone do not yield a universal Hilbert–Schmidt criterion. In the i.i.d. setting with (1), theorem 1.1 gives the precise Hilbert–Schmidt condition modulo $\mathcal{P}$ for every bounded invertible operator. The existence of non-Gaussian quasi-invariant measures on linear spaces [11, 38] addresses a different question from this classification for a fixed product law.

The discrete stabilizer is related to the classical characterization of Gaussian laws by independence of linear forms [9, 15, 16, 36] and to finite-dimensional identifiability in independent component analysis [8]. These results concern exact independence. Here the measure-class problem permits dependent output coordinates: the additional conclusion is square summability of the deviation from an allowed coordinate permutation, detected by the actual output marginals.

1.3 Two boundary regimes

The finite-second-moment hypothesis determines the quadratic accumulation law. For the standard symmetric α -stable product, $0 < \alpha < 2$, theorem 4.1 shows, within the compatible stable row-series class, that equivalence holds precisely when an allowed signed permutation reduces the operator to a matrix $M = (m_{ij})$ satisfying

$$\sum_i |m_{ii} - 1|^2 + \sum_j \left(\sum_{i \neq j} |m_{ij}|^2 \right)^{\alpha/2} < \infty.$$

The first term is quadratic, whereas the second is an $\ell^\alpha(\ell^2)$ column cost. Hence an everywhere-positive density and finite Fisher information do not by themselves make Hilbert–Schmidt proximity sufficient.

When a finite-variance density may vanish, the same Fourier/Fisher argument still gives Hilbert–Schmidt necessity. The exact extension in theorem 5.2 adds the condition that every row sum of both T and T^{-1} lies almost surely in $\{f > 0\}$. For bounded or one-sided coordinate laws, support geometry is stronger still: theorem 5.3 reduces the equivalence group to explicit signed-permutation or permutation classes without density or Fisher assumptions.

1.4 Proof strategy and organization

The finite-variance proof has three main steps. First, non-Gaussianity is encoded by the rotational Fourier defect

$$R(\xi, \eta) = \eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta), \quad \phi(u) = \mathbb{E}e^{iuX}.$$

The determinant of the two weighted Fourier jets $(\xi\phi(\xi), \phi'(\xi))$ and $(\eta\phi(\eta), \phi'(\eta))$ vanishes at every pair only for a Gaussian law. A finite set of frequencies then yields bounded tests for all matrix directions. Combined with the matrix Fisher operator, these tests give local Hellinger coercivity in Frobenius norm with constants independent of the dimension.

Second, the operator action must be realized on $\mathbb{R}^{\mathbb{N}}$, since a product sample need not belong to ℓ^2 . A common measurable carrier makes row series composable and allows inverse maps and

likelihood cocycles to be used without incompatible null sets. Translation ergodicity gives the equivalent–singular dichotomy and identifies bounded projected energy with equivalence.

Third, absolute continuity initially gives only asymptotic matching of remote rows with allowed signed coordinates. If the squared matching errors were not summable, tail-coordinate swaps would package a fixed amount of error into a small Hilbert–Schmidt perturbation. Local coercivity detects this perturbation, while absolute continuity makes the same remote swap asymptotically invisible. The contradiction proves Hilbert–Schmidt necessity.

[Section 2](#) develops the dimension-free Fourier and Hellinger geometry, and [Section 3](#) proves the finite-variance classification. The stable and zero-set/support regimes are treated in [Sections 4](#) and [5](#). We use A^* for the adjoint, P_n for coordinate projection on H , and $\|\cdot\|_F$ for the Frobenius norm.

2 Dimension-Free Hellinger Geometry

The rotational Fourier defect distinguishes every non-Gaussian coordinate law without dividing by its characteristic function. Fixed Fourier frequencies then produce bounded tests for all matrix directions, and the matrix Fisher operator identifies their infinitesimal cost. The goal of this section is the dimension-free local estimate in [proposition 2.6](#); the global estimate is derived from the classification in [Section 3.8](#).

2.1 A product of centered complex exponentials

This subsection requires only $\mathbb{E}X = 0$ and $\mathbb{E}|X| < \infty$. Neither a density nor a second moment is needed. Write $\phi(u) = \mathbb{E}e^{iuX}$, and for independent copies X_1, X_2 put

$$O_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad W_{\xi, \eta}(x, y) = (e^{i\xi x} - \phi(\xi))(e^{i\eta y} - \phi(\eta)).$$

This fixes the sign convention for rotation. Centering removes the two marginal responses. It leaves only the dependence created by rotation. In particular,

$$\mathbb{E}W_{\xi, \eta}(X_1, X_2) = 0, \quad \mathbb{E}|W_{\xi, \eta}(X_1, X_2)|^2 = (1 - |\phi(\xi)|^2)(1 - |\phi(\eta)|^2) \leq 1. \quad (6)$$

Pointwise, $|W_{\xi, \eta}| \leq 4$, so its variance under an arbitrary comparison law is at most 16. Both bounds are dimension-free and require no tail assumption.

2.2 Gaussian characterization by a global linear ODE

The next lemma makes the rotational Fourier defect a logarithm-free rigidity tool, valid even when ϕ has zeros. It is an infinitesimal version of the Gaussian rigidity behind independence characterizations [[15](#), [16](#)]: every non-Gaussian law has a nonzero rotational response.

Lemma 2.1 (Gaussian characterization by the rotational Fourier defect). *For the product of centered complex exponentials defined above,*

$$R(\xi, \eta) = \frac{d}{d\theta} \mathbb{E}W_{\xi, \eta}(O_\theta(X_1, X_2)) \Big|_{\theta=0} = \eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta). \quad (7)$$

Moreover,

$$R \equiv 0 \iff \text{Law}(X) \text{ is a centered Gaussian, possibly degenerate.} \quad (8)$$

Equivalently, all pairs of values of the weighted Fourier jet

$$u \mapsto (u\phi(u), \phi'(u))$$

are linearly dependent if and only if $\text{Law}(X)$ is a centered Gaussian, possibly degenerate.

Proof. Since $\mathbb{E}|X| < \infty$, we have $\phi \in C^1(\mathbb{R})$ and $\phi'(0) = i\mathbb{E}X = 0$. After rotation, the joint characteristic function is

$$\phi(\xi \cos \theta + \eta \sin \theta) \phi(-\xi \sin \theta + \eta \cos \theta).$$

The two marginal terms in the centered product have zero derivative at $\theta = 0$, again by $\phi'(0) = 0$; differentiating the displayed product gives (7).

Now suppose $R \equiv 0$. Choose $\eta_0 \neq 0$ with $\phi(\eta_0) \neq 0$, possible because $\phi(0) = 1$. Then, for every ξ ,

$$\phi'(\xi) = c\xi\phi(\xi), \quad c = \frac{\phi'(\eta_0)}{\eta_0\phi(\eta_0)}, \quad \phi(0) = 1.$$

Uniqueness for this global linear ODE yields $\phi(\xi) = e^{c\xi^2/2}$. The identities $\phi(-\xi) = \overline{\phi(\xi)}$ and $|\phi(\xi)| \leq 1$ imply respectively $c \in \mathbb{R}$ and $c \leq 0$. Thus $c = -\sigma^2$ for some $\sigma \geq 0$, so X is centered Gaussian (the case $\sigma = 0$ is the point mass at zero). Conversely, direct substitution shows that every such Gaussian has $R \equiv 0$. $\square$

2.3 The determinant representation and finite detection

The determinant form supplies the finite frequency pair used to construct the dual moments in Section 2.7.

Consider the curve

$$u \mapsto (u\phi(u), \phi'(u)) \in \mathbb{C}^2,$$

described as the weighted Fourier jet. Its values satisfy

$$\det \begin{pmatrix} \xi\phi(\xi) & \phi'(\xi) \\ \eta\phi(\eta) & \phi'(\eta) \end{pmatrix} = -R(\xi, \eta). \quad (9)$$

The weighting by u is essential: for a nondegenerate Gaussian law, the unweighted curve $u \mapsto (\phi(u), \phi'(u))$ is not contained in a fixed line.

For a non-Gaussian law, continuity supplies distinct frequencies and a compact detecting neighborhood, independent of the ambient dimension; the diagonal $\xi = \eta$ never detects the law.

The same product of centered complex exponentials gives a local Hellinger lower bound for one rotation. Choose a phase α with $\text{Re}(e^{-i\alpha}R(\xi, \eta)) > 0$ and apply

$$|\mathbb{E}_\lambda F - \mathbb{E}_\nu F| \leq h(\lambda, \nu) \{2\mathbb{E}_\lambda F^2 + 2\mathbb{E}_\nu F^2\}^{1/2}$$

to $F = \text{Re}(e^{-i\alpha}W_{\xi, \eta})$. It follows that $h((O_\theta)_\# \text{Law}(X)^{\otimes 2}, \text{Law}(X)^{\otimes 2}) \geq c|\theta|$ for small $|\theta|$. A first moment suffices for this lower bound. The dimension-free matrix estimates below require the additional second-moment and Sobolev hypotheses.

2.4 Dimension-free variance and affine normalization

The dimension-free variance bounds needed below are already contained in (6); we record only how R changes under affine normalization.

Centering, scaling, and phase. Let Y have mean m and characteristic function ϕ_Y , and put $X = (Y - m)/\sigma$, where $\sigma > 0$. For $a = \xi/\sigma$ and $b = \eta/\sigma$, the rotational Fourier defect of X satisfies

$$R(\xi, \eta) = e^{-i(m/\sigma)(\xi+\eta)} [b\phi'_Y(a)\phi_Y(b) - a\phi_Y(a)\phi'_Y(b) + im(a-b)\phi_Y(a)\phi_Y(b)]. \quad (10)$$

The bracketed term is the mean-corrected rotational response for two copies of Y . Centering contributes the displayed phase, scaling changes the frequencies, and the modulus at corresponding frequencies is unchanged. For a centered symmetric law the defect is real, but its phase at a single pair is not a complete measure of skewness.

For the concrete comparison $Y \sim \text{Gamma}(2, 1)$, $m = 2$, $\sigma = \sqrt{2}$, and $(\xi, \eta) = (1, 2)$, the expression in brackets in (10) equals $4/27$, so the law centered with variance one has

$$R(1, 2) = \frac{4}{27} e^{-3i\sqrt{2}}.$$

Thus the defect can remain explicit even for an asymmetric law with a support boundary.

2.5 The matrix Fisher operator

The matrix Fisher operator separates diagonal scale, symmetric off-diagonal, and skew-symmetric rotational directions. Its rotational eigenvalue is $J - 1$, which vanishes exactly in the Gaussian case. The general interpretation of Fisher information as a metric on density families is developed in [1]; here the matrix decomposition specifies the cost of each mixing direction.

Let f be a probability density and write $g = \sqrt{f}$. In this subsection we assume

$$\begin{aligned} f > 0 \quad \text{Lebesgue-almost everywhere,} \quad g &\in W^{1,2}(\mathbb{R}), \quad xg'(x) \in L^2(\mathbb{R}), \\ \mathbb{E}X &= 0, \quad \mathbb{E}X^2 = 1, \quad X \sim f(x) dx. \end{aligned} \quad (11)$$

The derivatives in these assumptions are weak derivatives. Define the location score $s = -f'/f = -2g'/g$ almost everywhere. Then

$$\mathbb{E}s(X) = 0, \quad \mathbb{E}[Xs(X)] = 1, \quad J := \mathbb{E}s(X)^2 = 4\|g'\|_2^2. \quad (12)$$

Define the scale score and its second moment by

$$d(x) := xs(x) - 1, \quad \kappa := \mathbb{E}d(X)^2 = 4\|xg'\|_2^2 - 1. \quad (13)$$

The two integration-by-parts identities in (12) follow from these assumptions. Indeed, $f' = 2gg'$ belongs to L^1 , and xf' belongs to L^1 by Cauchy–Schwarz and the finite second moment. Choose a smooth compactly supported cutoff χ equal to one near zero. Applying integration by parts first to $\chi(x/L)$ and then to $x\chi(x/L)$ gives the identities as $L \rightarrow \infty$.

For independent copies $X_1, \dots, X_n$ of X , write $\mathbf{X}^{(n)} = (X_1, \dots, X_n) \sim \mu_n$. For $A \in M_n(\mathbb{R})$, define the score of the pushforward by $I + tA$ at $t = 0$ by

$$S_A(\mathbf{X}^{(n)}) = \sum_{i,j=1}^n A_{ij}(s(X_i)X_j - \delta_{ij}) = \sum_i A_{ii}d(X_i) + \sum_{i \neq j} A_{ij}s(X_i)X_j. \quad (14)$$

Theorem 2.2 (The matrix Fisher operator). *Under (11), let s and d be the location and scale scores defined above, and define the matrix Fisher operator $\mathcal{I}$ by $\langle A, \mathcal{I}B \rangle_F = \mathbb{E}[S_A S_B]$. Then*

$$\boxed{\mathcal{I}A = JA + A^* + (\kappa - J - 1) \text{diag } A.} \quad (15)$$

Here $\text{diag } A$ is the matrix with diagonal entries A_{ii} and zero off-diagonal entries. The diagonal, symmetric off-diagonal, and skew-symmetric matrix subspaces are orthogonal for the Frobenius inner product and have eigenvalues κ , $J + 1$, and $J - 1$, respectively. Equivalently,

$$\begin{aligned} \langle A, \mathcal{I}A \rangle_F &= \kappa \|\text{diag } A\|_F^2 + (J + 1) \left\| \frac{A + A^*}{2} - \text{diag } A \right\|_F^2 \\ &\quad + (J - 1) \left\| \frac{A - A^*}{2} \right\|_F^2. \end{aligned} \quad (16)$$

The eigenvalue multiplicities are, respectively,

$$n, \quad n(n - 1)/2, \quad n(n - 1)/2. \quad (17)$$

Moreover, $\kappa > 0$ and $J \geq 1$. Here $J = 1$ holds precisely for the standard Gaussian density. Thus, for a non-Gaussian f , $\min\{\kappa, J - 1\} > 0$ and

$$\begin{aligned} \min\{\kappa, J - 1\} \|A\|_F^2 &\leq \langle A, \mathcal{I}A \rangle_F \leq C_f \|A\|_F^2, \\ C_f &:= \max\{\kappa, J + 1\}. \end{aligned} \quad (18)$$

Proof. The variables $d(X_i)$ are centered and independent. Their covariances with the off-diagonal terms vanish, while two off-diagonal terms have nonzero covariance only when their ordered index pairs coincide or are reversed; the corresponding covariances are J and $(\mathbb{E}Xs(X))^2 = 1$. Hence

$$\mathbb{E}S_A^2 = \kappa \sum_i A_{ii}^2 + \sum_{i < j} \{J(A_{ij}^2 + A_{ji}^2) + 2A_{ij}A_{ji}\}. \quad (19)$$

Polarization gives (15); resolving each off-diagonal pair into its symmetric and skew-symmetric parts gives (16) and the stated multiplicities.

Cauchy–Schwarz applied to $\mathbb{E}Xs(X) = 1$ gives $J \geq 1$. Equality implies $s(X) = X$ almost surely, hence $g' = -xg/2$ almost everywhere and f is the standard Gaussian. If $\kappa = 0$, then $2xg' + g = 0$ almost everywhere on each of the two half-lines. The absolutely continuous solutions there are constant multiples of $|x|^{-1/2}$, which cannot give a positive probability density. This proves $\kappa > 0$ and the asserted bounds. $\square$

2.6 Hellinger paths, polar decomposition, and symmetric perturbations

For orientation, write the left polar decomposition of an invertible matrix as $T = BO$, where $B = (TT^*)^{1/2}$ is positive and O is orthogonal. The covariance defect is $TT^* - I = B^2 - I$, so a finite second moment detects the symmetric factor. The skew tangent of the orthogonal factor has Fisher eigenvalue $J - 1$, which is positive precisely off the Gaussian law. The full matrix Fisher operator decomposition above packages these two mechanisms without requiring a nonlinear separation of B and O at every step.

The square-root formulation is the one used in quadratic-mean differentiability [39, Section 7.2]. Logarithmic derivatives and the relation between differentiation of measures and

quasi-invariance are treated in [3, 5]. For the present linear transformations, the half-density action below gives a path bound whose constant does not grow with the number of coordinates.

For $A \in \text{GL}(n, \mathbb{R})$ define a unitary operator on $L^2(\mathbb{R}^n, dx)$ by

$$(U_A v)(y) = |\det A|^{-1/2} v(A^{-1}y), \quad \Omega_n(x) = \prod_{i=1}^n g(x_i).$$

Thus $U_A \Omega_n$ is the square root of the Lebesgue density of $A_{\#} \mu_n$. For invertible A, B ,

$$h(A_{\#} \mu_n, B_{\#} \mu_n) = \|U_A \Omega_n - U_B \Omega_n\|_2.$$

For a fixed real matrix A , the generator of $U_{\exp(tA)}$ is

$$G_A v = -\frac{1}{2} \text{tr}(A) v - (Ax) \cdot \nabla v, \quad G_A \Omega_n = \frac{1}{2} S_A \Omega_n. \quad (20)$$

These identities hold in L^2 . To justify them under (11), observe that $x_j \partial_i \Omega_n \in L^2$ for all i, j : the diagonal case uses $xg' \in L^2$, and the off-diagonal case uses $g' \in L^2$ and $\mathbb{E}X^2 = 1$. Cutoff and mollification approximate Ω_n in the corresponding weighted Sobolev graph norm, where the usual change-of-variables differentiation is valid.

Lemma 2.3 (A dimension-free Hellinger path bound). *If A_t , $0 \leq t \leq 1$, is a continuously differentiable path in $\text{GL}(n, \mathbb{R})$, then $t \mapsto U_{A_t} \Omega_n$ is continuously differentiable in L^2 and*

$$\begin{aligned} \frac{d}{dt} U_{A_t} \Omega_n &= U_{A_t} G_{A_t^{-1} \dot{A}_t} \Omega_n, \\ \left\| \frac{d}{dt} U_{A_t} \Omega_n \right\|_2 &= \frac{1}{2} \sqrt{\left\langle A_t^{-1} \dot{A}_t, \mathcal{I}(A_t^{-1} \dot{A}_t) \right\rangle_F}. \end{aligned} \quad (21)$$

Consequently,

$$h((A_1)_{\#} \mu_n, (A_0)_{\#} \mu_n) \leq \frac{\sqrt{C_f}}{2} \int_0^1 \|A_t^{-1} \dot{A}_t\|_F dt. \quad (22)$$

In particular, if $\|B\|_{\text{op}} \leq r < 1$, then

$$h((I + B)_{\#} \mu_n, \mu_n) \leq \frac{\sqrt{C_f}}{2(1-r)} \|B\|_F. \quad (23)$$

All constants are independent of n .

Proof. The identity $U_A U_B = U_{AB}$ and (20) give the derivative formula. Unitarity and (18) give its norm; integration proves (22). For the last bound take $A_t = I + tB$ and use $\|A_t^{-1}\|_{\text{op}} \leq (1-r)^{-1}$. $\square$

2.7 Finite real Fourier sums with dual moments

Assume in addition that f is non-Gaussian. The score S_A is the correct tangent, but it is not a usable dimension-free test after the law moves: differentiating a score-based test would require regularity and integrability not assumed here. We instead need, for every matrix direction, a test built from bounded one-coordinate functions, with exact duality against the score and with L^2 -norm bounded by a constant times the Frobenius norm of that direction, uniformly in the dimension. The rank-two obstruction in lemma 2.1 produces the two off-diagonal coordinate functions; a separate small-frequency argument using the second moment produces the diagonal coordinate function. All three are finite real Fourier sums at frequencies depending on f but not on the matrix dimension.

Lemma 2.4 (Finite real Fourier sums with dual moments). *There exist centered finite real Fourier sums p, q, r such that*

$$\begin{array}{c|cc} & s & x \\ \hline p & 1 & 0 \\ q & 0 & 1 \end{array} \quad \text{and} \quad \mathbb{E}[r(X)d(X)] = 1. \quad (24)$$

The entries in the table denote expectations of the products. For

$$H_A(\mathbf{X}^{(n)}) := \sum_i A_{ii}r(X_i) + \sum_{i \neq j} A_{ij}p(X_i)q(X_j), \quad (25)$$

one has, for all real matrices A, B of the same size,

$$\mathbb{E}[H_A S_B] = \langle A, B \rangle_F, \quad \|H_A\|_{L^2(\mu_n)} \leq C_{\text{dual}} \|A\|_F, \quad (26)$$

where C_{dual} is independent of n .

Proof. On the real space of centered finite real Fourier sums, consider

$$L_1(u) = \mathbb{E}u'(X) = \mathbb{E}[u(X)s(X)], \quad L_2(u) = \mathbb{E}[Xu(X)].$$

The integration-by-parts identity is valid because u, u' are bounded. If $u \mapsto (L_1(u), L_2(u))$ had rank at most one, then after complexification its determinant on the centered exponentials of frequencies ξ, η would vanish. Since $L_1(e^{i\xi x}) = i\xi\phi(\xi)$ and $L_2(e^{i\xi x}) = -i\phi'(\xi)$, this would give

$$\eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta) = 0 \quad (\xi, \eta \in \mathbb{R}).$$

By lemma 2.1, this contradicts non-Gaussianity. Thus the moment map has rank two, and inverting the moment matrix of two Fourier sums gives p, q with the required dual moments.

For $\xi \neq 0$ let $r_0(x) = \cos(\xi x) - \mathbb{E}\cos(\xi X)$. Dominated convergence gives

$$\frac{\mathbb{E}[Xr'_0(X)]}{\xi^2} = -\mathbb{E}\left[X\frac{\sin(\xi X)}{\xi}\right] \longrightarrow -\mathbb{E}X^2 = -1 \quad (\xi \rightarrow 0).$$

For small nonzero ξ the denominator is nonzero, so $r = r_0/\mathbb{E}[Xr'_0(X)]$ satisfies, by cutoff integration by parts, $\mathbb{E}[r(X)d(X)] = \mathbb{E}[Xr'(X)] = 1$.

Centering kills all diagonal-off-diagonal cross terms. Coincident ordered pairs contribute $\mathbb{E}[ps]\mathbb{E}[qX] = 1$, whereas reversed pairs contribute $\mathbb{E}[pX]\mathbb{E}[qs] = 0$; hence $\mathbb{E}[H_A S_B] = \langle A, B \rangle_F$. Finally, independence gives, for centered $a, b \in L^2(f)$,

$$\left\| \sum_{i \neq j} A_{ij}a(X_i)b(X_j) \right\|_2 \leq \sqrt{2} \|a\|_2 \|b\|_2 \|A\|_F. \quad (27)$$

Combining this with orthogonality of the centered diagonal terms proves the uniform norm bound. $\square$

The test functions H_A will be paired with the Hellinger tangent along a matrix path. To keep that pairing uniformly controlled after moving away from the identity, we need one uniform derivative bound for $H_A \Omega_n$. A naive term count grows with n , while a direct Hölder estimate asks for integrability that is unavailable. The next lemma isolates this non-formal step. Its proof groups terms by their index-coincidence patterns, turning every surviving coefficient array into a Frobenius-controlled contraction. In particular, it uses no fourth moment of X or higher moment of the score.

Lemma 2.5 (Generator bound for the functions H_A). *Let p, q, r be any fixed centered bounded continuously differentiable functions with bounded derivatives. Define H_A by (25). There is a constant C_{gen} , depending on these functions and on J, κ , such that*

$$\|G_B(H_A\Omega_n)\|_{L^2(dx)} \leq C_{\text{gen}}\|A\|_F\|B\|_F \quad (A, B \in M_n(\mathbb{R}), n \geq 1). \quad (28)$$

Proof. For fixed n , both H_A and its first derivatives are bounded. Consequently

$$x_j \partial_i (H_A \Omega_n) = H_A x_j \partial_i \Omega_n + x_j (\partial_i H_A) \Omega_n \in L^2(dx)$$

by the weighted Sobolev regularity of Ω_n and $\mathbb{E}X^2 = 1$. The cutoff and mollification argument used for (20) therefore puts $H_A \Omega_n$ in the domain of G_B . The product rule gives

$$G_B(H_A \Omega_n) = \left\{ \frac{1}{2} H_A S_B - (Bx) \cdot \nabla H_A \right\} \Omega_n. \quad (29)$$

We bound the two terms uniformly in n by grouping indices according to their coincidences.

We will repeatedly use the following elementary consequence of independence. If $u_1, \dots, u_m$ are centered $L^2(f)$ functions, and a coefficient tensor C is indexed by ordered tuples of *distinct* coordinates, then

$$\left\| \sum_{i_1, \dots, i_m}^{\neq} C_{i_1 \dots i_m} \prod_{a=1}^m u_a(X_{i_a}) \right\|_2 \leq \sqrt{m!} \|C\|_{\ell^2} \prod_{a=1}^m \|u_a\|_2. \quad (30)$$

Indeed, after squaring, a mixed term vanishes unless the two tuples have the same index set; summing the at most $m!$ permutations and applying Cauchy–Schwarz proves the estimate. We use it only for $m \leq 4$.

First consider $H_A S_B$. For a vector a and a zero-diagonal matrix M , abbreviate

$$D_a(u) = \sum_i a_i u(X_i), \quad Q_M(u, v) = \sum_{i \neq j} M_{ij} u(X_i) v(X_j).$$

The product $H_A S_B$ is the sum of the four products obtained by multiplying $D_{\mathbf{a}}(r) + Q_{A_0}(p, q)$ and $D_{\mathbf{b}}(d) + Q_{B_0}(s, x)$, where $\mathbf{a} = (A_{ii})_i$, $\mathbf{b} = (B_{ii})_i$, $A_0 = A - \text{diag } A$, and $B_0 = B - \text{diag } B$. Every local product on a shared coordinate is a bounded function times one of d, s, x , hence lies in $L^2(f)$; split it into its mean and centered part. For $D_a(u) D_b(v)$, the distinct-index tensor, diagonal Hadamard vector, and total contraction all have norm at most $\|a\|_2 \|b\|_2$. For $D_a(u) Q_M(v, w)$, the three coincidence patterns give $a \otimes M$, a row or column multiplication of M by a , and Ma or M^*a ; all have norm at most $\|a\|_2 \|M\|_F$. The reversed product is identical.

It remains only to record the contractions for

$$Q_M(u, v) Q_N(w, z) = \sum_{i \neq j, k \neq l} M_{ij} N_{kl} u_i v_j w_k z_l,$$

where u, v are bounded and $w, z \in L^2$. Four distinct indices give $M \otimes N$. For the representative one-coincidence pattern $i = k$, the centered local product has coefficient norm

$$\left\{ \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_l N_{il}^2 \right) \right\}^{1/2} \leq \|M\|_F \|N\|_F.$$

Its mean contraction is the off-diagonal part of M^*N ; the other three one-coincidence patterns are transposes of this one. With both indices shared, the coefficients are $M_{ij}N_{ij}$ or $M_{ij}N_{ji}$ and their row, column, or total contractions. For example,

$$\sum_i \left| \sum_j M_{ij}N_{ij} \right|^2 \leq \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_j N_{ij}^2 \right) \leq \|M\|_F^2 \|N\|_F^2,$$

and Cauchy–Schwarz gives the same bound for every transpose, Hadamard, and total contraction. These patterns exhaust all coincidences. Applying (30) to the centered pieces proves

$$\|H_A S_B\|_2 \leq C \|A\|_F \|B\|_F. \quad (31)$$

For $(Bx) \cdot \nabla H_A$, the diagonal part is

$$\sum_{i,k} A_{ii} B_{ik} r'(X_i) X_k.$$

For $i \neq k$, centering r' produces the matrix $(\text{diag } A)B$ and the vector $B^* \mathbf{a} - \mathbf{a} \odot \mathbf{b}$; for $i = k$, center $xr'(x)$. Their coefficient norms are at most $2\|A\|_F \|B\|_F$.

Let $M = A_0$. The derivative of the off-diagonal part is the sum of

$$F(M, B; p', q) := \sum_{i \neq j, k} M_{ij} B_{ik} p'(X_i) q(X_j) X_k$$

and $F(M^*, B; q', p)$. For $k \notin \{i, j\}$, centering p' gives a three-index coefficient of norm at most

$$\left\{ \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_k B_{ik}^2 \right) \right\}^{1/2} \leq \|M\|_F \|B\|_F.$$

and the mean gives $M^*B - M^* \text{diag } B$, of norm at most $2\|M\|_F \|B\|_F$. For $k = i$, centering xp' gives the matrix $(B_{ii}M_{ij})$ and vector $M^* \mathbf{b}$; for $k = j$, the contractions are $M \odot B$ and its row, column, and total sums. The preceding Hadamard estimates bound all of them by $\|M\|_F \|B\|_F$. The local functions lie in L^2 because p', q are bounded and $\mathbb{E}X^2 = 1$. The same argument applies to $F(M^*, B; q', p)$, so (30) yields

$$\|(Bx) \cdot \nabla H_A\|_2 \leq C \|A\|_F \|B\|_F.$$

Together with (31) and (29), this proves the lemma. $\square$

We can now close the local argument. The test functions H_A give a linear functional that sees the first-order displacement, while the generator bound controls the error made along the exponential path. This turns the nonzero rotational Fourier defect into a uniform Hellinger lower bound for every small matrix perturbation.

Proposition 2.6 (Uniform local Hellinger coercivity). *For every non-Gaussian density satisfying (11), there are constants $c, C, \delta > 0$, independent of n , such that*

$$c\|B\|_F \leq h((I + B)_{\#} \mu_n, \mu_n) \leq C\|B\|_F \quad \text{whenever } \|B\|_F \leq \delta. \quad (32)$$

The lower bound can be obtained using the fixed finite set of coordinate frequencies from lemma 2.4.

Proof. Fix the finite real Fourier sums from [lemma 2.4](#). In exponential coordinates, unitarity and the group derivative give

$$\langle H_A \Omega_n, (U_{\exp A} - I) \Omega_n \rangle = \int_0^1 \langle U_{\exp(-tA)}(H_A \Omega_n), G_A \Omega_n \rangle dt.$$

The value obtained by replacing $U_{\exp(-tA)}(H_A \Omega_n)$ with $H_A \Omega_n$ is $\frac{1}{2} \mathbb{E} H_A S_A = \frac{1}{2} \|A\|_F^2$. Moreover, $\|(U_{\exp(-tA)} - I)v\|_2 \leq t \|G_A v\|_2$ on the generator domain. Therefore [lemma 2.5](#) and (18) yield

$$\left| \langle H_A \Omega_n, (U_{\exp A} - I) \Omega_n \rangle - \frac{1}{2} \|A\|_F^2 \right| \leq \frac{C_{\text{gen}} \sqrt{C_f}}{4} \|A\|_F^3.$$

For sufficiently small $\|A\|_F$, Cauchy–Schwarz and (26) thus give

$$h((\exp A)_{\#} \mu_n, \mu_n) \geq \frac{1}{4C_{\text{dual}}} \|A\|_F.$$

For $\|B\|_{\text{op}} < 1/2$, the real power-series logarithm $A = \log(I + B)$ exists and satisfies

$$\|A - B\|_F \leq \sum_{k \geq 2} \frac{\|B\|_{\text{op}}^{k-1}}{k} \|B\|_F \leq C \|B\|_{\text{op}} \|B\|_F.$$

Thus $\|A\|_F$ and $\|B\|_F$ are uniformly comparable in a fixed small chart, proving the lower bound. The upper bound is [lemma 2.3](#). $\square$

The half-density pairing supplies the uniform local certificate; tail symmetry in [Section 3.5](#) applies it to arbitrary output marginals without a separate mixed-law variance bound for H_A .

2.8 Uniform coercivity modulo the exact linear stabilizer

The following global form is proved in [Section 3.8](#), after the infinite-dimensional classification supplies the compactness input; it is not used in that classification.

Theorem 2.7 (Dimension-free global Hellinger coercivity). *Assume (1). Let $\mathcal{P}_n$ denote the allowed signed-permutation matrices in dimension n , and set*

$$d_n(T) = \min_{Q \in \mathcal{P}_n} \|T - Q\|_F. \quad (33)$$

For every $0 < a \leq b < \infty$ there are constants $c = c(f, a, b) > 0$ and $C = C(f, a, b) < \infty$ such that, for every n and every $T \in \text{GL}(n, \mathbb{R})$ whose singular values lie in $[a, b]$,

$$c \min\{d_n(T)^2, 1\} \leq h(\mu_n, T_{\#} \mu_n)^2 \leq C \min\{d_n(T)^2, 1\}. \quad (34)$$

3 The Finite-Variance Classification

We first realize row series on a common measurable carrier and prove the equivalent–singular orbit dichotomy. Hilbert–Schmidt sufficiency follows by finite-coordinate approximation. For necessity, qualitative tail rigidity is upgraded to square summability by conjugating remote coordinate swaps and applying the local estimate from [proposition 2.6](#).

3.1 Measurable row-series realizations on countable operator orbits

Only $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$ are needed here. Each row series exists as an L^2 equivalence class. The distinction between an operator on ℓ^2 and its measurable realization on $\mathbb{R}^{\mathbb{N}}$ is already central to [34]. Such equivalence classes cannot be composed pointwise: separately chosen versions for different operators or orbit laws may fail on incompatible null sets. We therefore select deterministic subsequences for every row in a fixed countable operator group, so that operator multiplication becomes composition on one common measurable carrier. Countability is used only to intersect the exceptional sets; no universal pointwise action of all of $\mathrm{GL}(H)$ is asserted.

Lemma 3.1 (Measurable row-series realization on each countable operator orbit). *For every $T \in \mathcal{B}(H)$, there is a measurable row-series realization, namely a Borel map $\Phi_T : \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}^{\mathbb{N}}$ whose coordinates are the $L^2(\mu)$ row sums*

$$(\Phi_T x)_i = \sum_{j \geq 1} T_{ij} x_j.$$

These maps can be chosen so that the following statements hold. If a probability measure λ satisfies, for some finite $C_\lambda \geq 0$,

$$\int \left| \sum_j a_j x_j \right|^2 d\lambda(x) \leq C_\lambda \|a\|_2^2 \quad \text{for every finitely supported } a, \quad (35)$$

then $(\Phi_T)_\# \lambda$ also satisfies this condition. Its constant is at most $C_\lambda \|T\|_{\mathrm{op}}^2$. Also

$$\Phi_S \circ \Phi_T = \Phi_{ST} \quad \lambda\text{-almost surely} \quad (S, T \in \mathcal{B}(H)). \quad (36)$$

For each fixed countable subgroup $\Gamma \leq \mathrm{GL}(H)$, there is a Γ -invariant Borel set $D \subset \mathbb{R}^{\mathbb{N}}$ of full measure for every λ satisfying (35). On this set, these restrictions form an action by bimeasurable automorphisms. In particular, this applies to μ and all its Γ -orbit laws, including singular orbit laws.

Proof. For $a \in \ell^2$, the finite sums $\sum_{j \leq n} a_j x_j$ have a unique $L^2(\lambda)$ limit L_a . Choose

$$n_k(a) = \min \left\{ n \geq k : \sum_{j > n} |a_j|^2 \leq 2^{-4k} \right\}.$$

Define $\hat{L}_a(x)$ to be the limit of $\sum_{j \leq n_k(a)} a_j x_j$ when it exists, and zero otherwise. By (35),

$$\left\| \sum_{j \leq n_k(a)} a_j x_j - L_a \right\|_{L^2(\lambda)}^2 \leq C_\lambda 2^{-4k}$$

λ -almost surely. This deterministic definition works for every law satisfying (35). Set $\Phi_T(x)_i = \hat{L}_{T^* e_i}(x)$; the construction is Borel.

For finitely supported b , $\sum_i b_i \Phi_T(x)_i = L_{T^* b}$ almost surely, proving the asserted second-moment bound for the pushforward. For the i -th row of S ,

$$\sum_{j \leq n} S_{ij} \hat{L}_{T^* e_j} = L_{T^* P_n S^* e_i} \longrightarrow L_{T^* S^* e_i} \quad \text{in } L^2(\lambda).$$

The selected subsequence on the left also converges almost surely to $\Phi_S(\Phi_T(x))_i$, because the output law satisfies the same type of bound. Uniqueness of the limit proves (36).

For a countable Γ , let B be the Borel set on which all its row limits, identity relations, and pairwise composition relations hold. The preceding argument gives $\lambda(B) = 1$ for every admissible law. Put

$$D = \bigcap_{T \in \Gamma} \Phi_T^{-1}(B).$$

Every orbit law satisfies (35), so D has full measure. The composition identities make D invariant and $\Phi_{T^{-1}}$ the inverse of Φ_T on D . $\square$

Write $\nu_T = (\Phi_T)_\# \mu$. Every countable operator family used below therefore acts on one common measurable carrier. No universal pointwise action of all $\text{GL}(H)$ is needed; on the original product law these versions agree with the usual almost-sure row sums.

3.2 Rational translation invariance and the zero–one dichotomy

Quasi-invariance under ℓ^2 -translations eliminates the intermediate case: every bounded invertible orbit is either equivalent to μ or singular to it. The translation input belongs to the Shepp-space problem of identifying all shifts that preserve a product measure class [32]. Beyond scalar translations, this question extends to infinite products of commutative Lie groups and to coordinatewise group actions [14, 29]. Within the translation problem, finer questions concern quasi-invariance in all ℓ^q directions, $1 \leq q < 2$ [2], and the linearity and sequence-space description of the Shepp space [13]. Here its ℓ^2 quasi-invariance is combined with translation ergodicity; the following proof specializes the equivalent–singular mechanism of [27] to each bounded invertible orbit.

Lemma 3.2 (Dichotomy for every bounded invertible orbit). *For every $T \in \text{GL}(H)$,*

$$\nu_T \sim \mu \quad \text{or} \quad \nu_T \perp \mu. \quad (37)$$

This conclusion uses only the a.e. positivity of f , its centered finite second moment, and $\sqrt{f} \in W^{1,2}(\mathbb{R})$. In particular, it holds under (1).

Proof. Put $g = \sqrt{f}$. The translation estimate

$$1 - \mathbf{A}(f, f(\cdot - a)) = \frac{1}{2} \|g(\cdot - a) - g\|_2^2 \leq \frac{1}{2} \|g'\|_2^2 a^2$$

and Kakutani’s theorem imply $(x \mapsto x + a)_\# \mu \sim \mu$ for every $a \in \ell^2$. Since the scalar pairing of a row of T with $T^{-1}a$ is absolutely convergent,

$$\Phi_T(x + T^{-1}a) = \Phi_T(x) + a \quad \mu\text{-almost surely};$$

hence ν_T is also quasi-invariant under every ℓ^2 translation.

Let $\mathcal{A}$ be the countable additive group generated by $\{qe_j, qTe_j : q \in \mathbb{Q}, j \geq 1\}$. Both μ and ν_T are quasi-invariant and ergodic under translation by $\mathcal{A}$. Indeed, rational-translation invariance and positivity of the finite-dimensional densities make every finite-coordinate section constant almost everywhere (convolve with a smooth approximate identity); the event is therefore a tail event, so Kolmogorov’s zero–one law applies. Pullback by Φ_T proves the same assertion for ν_T , because translation by qTe_j pulls back to translation by qe_j .

Write $\nu_T = \nu_a + \nu_s$ for its Lebesgue decomposition relative to μ . Choose a Borel μ -null set N_0 supporting ν_s , and let $N = \bigcup_{a \in \mathcal{A}} (N_0 + a)$. Quasi-invariance of μ makes N a μ -null set; it is exactly $\mathcal{A}$ -invariant and still supports ν_s . Thus $\nu_T(N) = \nu_s(\mathbb{R}^N) \in \{0, 1\}$ by ergodicity. Mass one gives singularity. If the mass is zero, write $\nu_T = \ell_T \mu$. Quasi-invariance makes $\{\ell_T > 0\}$ invariant modulo μ ; it has positive measure and hence, by ergodicity, full measure. Thus $\nu_T \sim \mu$. $\square$

3.3 Finite-marginal likelihood ratios and projected Hellinger energy

Finite-marginal likelihood ratios form a martingale, but its limit is the density of the absolutely continuous part of ν_T , not automatically a density of the whole law. Martingale convergence therefore identifies the limiting affinity; the orbit dichotomy is the additional input that upgrades positive limiting affinity from mere overlap to equivalence. This is why bounded projected Hellinger energy is an exact diagnostic in the present orbit problem. For background on likelihood ratios and Hellinger comparison, see [19]; the role of Lebesgue decomposition in equivalence and singularity criteria is discussed in [22].

The convolution argument below establishes finite-dimensional absolute continuity. Global equivalence after an independent symmetric additive perturbation is a distinct problem studied in [18], under a second-derivative integrability condition on the coordinate density. For the dependent image law here, the global step uses translation ergodicity.

Lemma 3.3 (Projected Hellinger energy and finite-marginal likelihood ratios). *Assume $f > 0$ almost everywhere, $\sqrt{f} \in W^{1,2}(\mathbb{R})$, and $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$. For $T \in \text{GL}(H)$, every finite output law $(\pi_n)_\# \nu_T$ is equivalent to μ_n . Writing $\mathcal{F}_n = \sigma(x_1, \dots, x_n)$, define*

$$\begin{aligned} \ell_{T,n}(x) &= \frac{d[(\pi_n)_\# \nu_T]}{d\mu_n}(\pi_n x), & E_n(T) &= -2 \log \mathbb{E}_\mu \sqrt{\ell_{T,n}}, \\ A(\mu_n, (\pi_n)_\# \nu_T) &= \mathbb{E}_\mu \sqrt{\ell_{T,n}}. \end{aligned} \quad (38)$$

Then $(\ell_{T,n})_{n \geq 1}$ is a nonnegative mean-one μ -martingale. We have

$$A(\mu_n, (\pi_n)_\# \nu_T) \downarrow A(\mu, \nu_T), \quad \nu_T \sim \mu \iff \sup_n E_n(T) < \infty. \quad (39)$$

Otherwise $E_n(T) \uparrow \infty$. Then $\nu_T \perp \mu$. If $\nu_T \sim \mu$ with density $\ell_T = d\nu_T/d\mu$, then

$$\ell_{T,n} = \mathbb{E}_\mu[\ell_T \mid \mathcal{F}_n] \longrightarrow \ell_T \quad \text{in } L^1(\mu), \quad (40)$$

$$\ell_{T,n}^{-1} = \mathbb{E}_{\nu_T}[\ell_T^{-1} \mid \mathcal{F}_n] \longrightarrow \ell_T^{-1} \quad \text{in } L^1(\nu_T). \quad (41)$$

In particular, both likelihood-ratio martingales are uniformly integrable in their respective reference measures.

Proof. The first n rows of T are linearly independent, since T^* is injective, so some n input columns form an invertible minor A . Writing the first n outputs as $AU + W$, with $U \sim \mu_n$ independent of W , shows that their density is a mixture of translates of the a.e.-positive density of AU . It is positive and finite almost everywhere, proving finite-dimensional equivalence.

Consistency of projected laws proves the martingale assertion. The martingale limit is $\ell_{T,\infty} = d(\nu_T)_a/d\mu$, the density of the absolutely continuous part of ν_T . Indeed, Fatou on cylinder sets gives $\ell_{T,\infty} \mu \leq \nu_T$; conversely, if $(\nu_T)_a = \ell_{ac} \mu$, then $\mathbb{E}_\mu[\ell_{ac} \mid \mathcal{F}_n] \leq \ell_{T,n}$, and martingale convergence gives the reverse inequality. Since $\mathbb{E}_\mu(\sqrt{\ell_{T,n}})^2 = 1$, the square roots are uniformly

integrable and converge in L^1 to $\sqrt{\ell_{T,\infty}}$. Conditional Jensen also makes their expectations decreasing; hence

$$\lim_n \mathbf{A}(\mu_n, (\pi_n)_\# \nu_T) = \int \sqrt{\ell_{T,\infty}} d\mu = \mathbf{A}(\mu, \nu_T).$$

The dichotomy in [lemma 3.2](#) now proves (39) and the singular alternative.

Under equivalence, the forward identity is the defining conditional-expectation identity for a marginal density. For the reverse, for $B \in \mathcal{F}_n$,

$$\int_B \ell_{T,n}^{-1} d\nu_T = \mu(B) = \int_B \ell_T^{-1} d\nu_T.$$

Conditional-expectation convergence gives both L^1 limits and hence both uniform-integrability assertions. $\square$

Because initial segments are cofinal among finite coordinate sets, data processing also shows that the criterion is independent of the coordinate enumeration. Here data processing is the contraction of Hellinger divergence under measurable projections, a special case of the divergence inequalities in [21].

3.4 Sufficiency of Hilbert–Schmidt perturbations

Finite-coordinate Hilbert–Schmidt approximants are Cauchy in Hellinger distance, but this alone produces only a candidate density and one direction of absolute continuity. The inverse approximants and the common measurable carrier are what prevent the limit from losing mass in the reverse direction.

Proposition 3.4 (Hilbert–Schmidt sufficiency and likelihood-ratio convergence). *Assume (11), let $\mu = \rho^{\otimes \mathbb{N}}$, where $\rho(dx) = f(x) dx$, and let $T = I + K \in \text{GL}(H)$ with $K \in \mathcal{S}_2$. Then $\nu_T \sim \mu$. For the eventually invertible operators $T_n = I + P_n K P_n$, their likelihood ratios $\ell_{T_n} = d\nu_{T_n}/d\mu$ converge in $L^1(\mu)$ to the strictly positive likelihood ratio $\ell_T = d\nu_T/d\mu$, and $\sqrt{\ell_{T_n}} \rightarrow \sqrt{\ell_T}$ in $L^2(\mu)$. The inverse approximants satisfy the same conclusion:*

$$\ell_{T_n^{-1}} \longrightarrow \ell_{T^{-1}} \quad \text{in } L^1(\mu).$$

For $S, T \in \text{GL}(H)$ with $S - I, T - I \in \mathcal{S}_2$, the likelihood ratios obey, on the common measurable carrier of their countable operator group,

$$\ell_{ST}(y) = \ell_S(y) \ell_T(\Phi_{S^{-1}} y), \tag{42}$$

$$\ell_T(y) \ell_{T^{-1}}(\Phi_{T^{-1}} y) = 1. \tag{43}$$

If Q is an exact linear stabilizer of μ , the same equivalence holds whenever $TQ^{-1} - I \in \mathcal{S}_2$ and T is invertible.

Proof. Since $P_n K P_n \rightarrow K$ in Hilbert–Schmidt norm, $T_n \rightarrow T$ in operator norm, so the approximants are eventually invertible with uniformly bounded inverse norms; their finite-coordinate likelihood ratios ℓ_{T_n} are positive almost everywhere. For all sufficiently large m, n , the operator $T_n^{-1} T_m - I$ acts in a finite coordinate block and has operator norm at most $1/2$. Hellinger invariance under the finite invertible map T_n^{-1} and (23) give

$$\|\sqrt{\ell_{T_m}} - \sqrt{\ell_{T_n}}\|_2 \leq \sqrt{C_f} \|T_n^{-1}\|_{\text{op}} \|T_m - T_n\|_{\mathcal{S}_2}. \tag{44}$$

Thus $\sqrt{\ell_{T_n}}$ converges in $L^2(\mu)$ to a nonnegative function of norm one; denote its square by $\tilde{\ell}_T$. Since

$$\|\ell_{T_m} - \ell_{T_n}\|_1 \leq 2\|\sqrt{\ell_{T_m}} - \sqrt{\ell_{T_n}}\|_2,$$

we also have $\ell_{T_n} \rightarrow \tilde{\ell}_T$ in L^1 . Meanwhile, $\Phi_{T_n}\mathbf{X} \rightarrow \Phi_T\mathbf{X}$ in L^2 on every finite set of output coordinates. Cylinder tests therefore identify $\ell_T\mu$ with ν_T .

The inverse approximants converge in Hilbert–Schmidt norm because

$$\|T_n^{-1} - T^{-1}\|_{\mathcal{S}_2} \leq \|T_n^{-1}\|_{\text{op}} \|T_n - T\|_{\mathcal{S}_2} \|T^{-1}\|_{\text{op}}.$$

Applying the same argument gives $\nu_{T^{-1}} \ll \mu$. On the common measurable carrier supplied by lemma 3.1,

$$\mu = (\Phi_T)_{\#}\nu_{T^{-1}} \ll (\Phi_T)_{\#}\mu = \nu_T.$$

This proves equivalence and identifies $\tilde{\ell}_T = \ell_T > 0$ almost everywhere. Change of variables on that carrier proves (42) and (43). Finally, $\nu_Q = \mu$ and $T = (TQ^{-1})Q$ give the assertion about exact stabilizers. $\square$

3.4.1 The orthogonal case requires only location Fisher information

For orthogonal transformations the assumptions can be reduced to

$$f > 0 \text{ almost everywhere, } g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \quad \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1. \quad (45)$$

In particular, $xg' \in L^2$ is not required. The cutoff identities $\mathbb{E}s(X) = 0$ and $\mathbb{E}[Xs(X)] = 1$ still follow from $J < \infty$ and the second moment. For a skew-symmetric matrix A , the score has only off-diagonal terms and

$$\mathbb{E}S_A^2 = (J - 1)\|A\|_F^2. \quad (46)$$

The generator formula remains valid. Only $x_j\partial_i\Omega_n$ with $i \neq j$ occur. One can justify it by radial cutoff and radial mollification. Those operators commute with the orthogonal one-parameter group.

Proposition 3.5 (Orthogonal local bounds and Hilbert–Schmidt sufficiency). *Under (45), every orthogonal U on ℓ^2 with $U - I \in \mathcal{S}_2$ satisfies $\nu_U \sim \mu$. Its likelihood ratios have the same L^1 and square-root L^2 convergence along finite-coordinate orthogonal approximants, as do the inverse likelihood ratios. If f is non-Gaussian, there also exist dimension-free $c, C, \delta > 0$ such that*

$$c\|V - I\|_F \leq h(V_{\#}\mu_n, \mu_n) \leq C\|V - I\|_F \quad (V \in O(n), \|V - I\|_F \leq \delta). \quad (47)$$

Proof. For a skew-symmetric A , integration along $\exp(tA)$ gives

$$h((\exp A)_{\#}\mu_n, \mu_n) \leq \frac{1}{2}\sqrt{J-1}\|A\|_F. \quad (48)$$

The construction of p, q in lemma 2.4 uses only $s(X), X \in L^2$ and $J - 1 > 0$. For skew-symmetric A , take $H_A = \sum_{i \neq j} A_{ij}p(X_i)q(X_j)$. Both H_A and S_A have no diagonal term, so the generator proof uses only bounded functions times $s(X)$ or X and remains valid under (45). The half-density pairing gives the local lower bound in exponential coordinates. Since a sufficiently small orthogonal $V - I$ has a real skew-symmetric logarithm with $\|\log V\|_F \asymp \|V - I\|_F$, this proves (47).

For the infinite-dimensional assertion, set $T_n = I + P_n(U - I)P_n$ and take its polar factor

$$V_n = T_n(T_n^* T_n)^{-1/2}.$$

It is orthogonal and acts identically outside the first n coordinates. Since $T_n \rightarrow U$ in Hilbert–Schmidt norm and its least singular value is eventually bounded below by some $a > 0$,

$$\|V_n - T_n\|_{\mathcal{S}_2} = \|I - |T_n|\|_{\mathcal{S}_2} \leq \frac{\|T_n^* T_n - I\|_{\mathcal{S}_2}}{1 + a}.$$

The right side tends to zero, so $V_n \rightarrow U$ in Hilbert–Schmidt norm. For large m, n , apply (48) to the small skew-symmetric logarithm of $V_n^* V_m$. The square roots of the likelihood ratios are Cauchy, and cylinder tests identify their limit as in [proposition 3.4](#). Repeating the argument for $V_n^{-1} \rightarrow U^{-1}$ and using the measurable inverse action proves equivalence and all asserted convergence. $\square$

3.5 Necessity of Hilbert–Schmidt perturbations

Approximation and likelihood-ratio convergence transfer local coercivity from coordinate blocks to Hilbert–Schmidt operators, allowing many small tail mismatches to be combined into a contradiction.

Lemma 3.6 (The local bound for Hilbert–Schmidt operators). *Let f be non-Gaussian and satisfy (11). There are constants $c, C, \delta > 0$ such that every $K \in \mathcal{S}_2(\ell^2)$ with $\|K\|_{\mathcal{S}_2} \leq \delta$ satisfies*

$$c\|K\|_{\mathcal{S}_2} \leq h(\nu_{I+K}, \mu) \leq C\|K\|_{\mathcal{S}_2}. \quad (49)$$

The constants can be chosen with $\delta < 1$, so $I + K$ is automatically invertible. Under the weaker assumptions (45), the same conclusion holds for orthogonal $I + K$ whenever f is non-Gaussian. This includes finite-rank perturbations whose ranges have infinite coordinate support.

Proof. Let $K_n = P_n K P_n$. The untouched tail has affinity one, so (32) applies directly to $h(\nu_{I+K_n}, \mu)$. By [proposition 3.4](#),

$$h(\nu_{I+K_n}, \mu) \longrightarrow h(\nu_{I+K}, \mu), \quad \|K_n\|_{\mathcal{S}_2} \longrightarrow \|K\|_{\mathcal{S}_2}.$$

Passing to the limit proves both bounds.

In the orthogonal case, the polar approximants from [proposition 3.5](#) satisfy

$$h(\nu_{V_n}, \mu) \longrightarrow h(\nu_{I+K}, \mu), \quad \|V_n - I\|_{\mathcal{S}_2} \longrightarrow \|K\|_{\mathcal{S}_2}.$$

After reducing the radius if necessary, pass to the limit in (47). $\square$

3.5.1 Characteristic-function rigidity and bounded invertible linear mixing

Necessity begins by identifying what a single output row can look like if its law is close to the original coordinate law. This is a one-dimensional rigidity problem for linear combinations of independent copies of X . The characteristic-function argument below says that, for a non-Gaussian coordinate law, such a row must asymptotically be one signed coordinate.

Throughout this subsection, X_j are independent copies of a real random variable X with law ρ and

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1,$$

and ρ is non-Gaussian. The characteristic-function arguments below require no moments of order greater than two. Recall

$$\phi(t) = \mathbb{E}e^{itX}, \quad \mu = \rho^{\otimes \mathbb{N}}.$$

The group $\mathcal{P}$ consists of coordinate permutations whose row signs preserve ρ . This definition applies even when ρ has no density.

Lemma 3.7 (A product approximation for small coefficients). *There is a nonnegative function ω_ρ , with $\omega_\rho(\delta) \rightarrow 0$ as $\delta \downarrow 0$, such that, for every finite or countable real family (b_j) with $\sum_j b_j^2 < \infty$,*

$$\left| \prod_j \phi(tb_j) - \exp\left(-\frac{t^2}{2} \sum_j b_j^2\right) \right| \leq \omega_\rho(\delta) t^2 \sum_j b_j^2 \quad \text{if} \quad \sup_j |tb_j| \leq \delta. \quad (50)$$

For a countable family, the product denotes the characteristic function of the L^2 sum $\sum_j b_j X_j$.

Proof. The centered second-moment assumption gives

$$\phi(u) = 1 - \frac{u^2}{2} + o(u^2).$$

Consequently

$$\omega_\rho(\delta) := \sup_{0 < |u| \leq \delta} \frac{|\phi(u) - e^{-u^2/2}|}{u^2} \longrightarrow 0, \quad \omega_\rho(0) := 0.$$

Both factors have modulus at most one. Telescoping finite products therefore bounds their difference by $\sum_j \omega_\rho(\delta) t^2 b_j^2$, proving (50). The countable case follows from L^2 convergence of the partial sums. $\square$

Lemma 3.8 (Qualitative rigidity of a unit row). *Let $a^{(n)} \in \ell^2$ satisfy $\|a^{(n)}\|_2 = 1$. If*

$$\text{Law}\left(\sum_j a_j^{(n)} X_j\right) \Longrightarrow \rho,$$

then

$$\inf_{\substack{j \geq 1, \varsigma \in \{-1, 1\} \\ \text{Law}(\varsigma X) = \rho}} \|a^{(n)} - \varsigma e_j\|_2 \longrightarrow 0. \quad (51)$$

Proof. Reorder each row by decreasing absolute value; this preserves the law of the sum and gives $|a_j^{(n)}| \leq j^{-1/2}$. From any subsequence, extract a further one with $a_j^{(n)} \rightarrow a_j$ for every j . Fatou gives

$$\tau_a := \sum_j a_j^2 \leq 1.$$

For fixed m, t , lemma 3.7 replaces the tail characteristic function by

$$\exp\left[-\frac{t^2}{2} \left(1 - \sum_{j \leq m} (a_j^{(n)})^2\right)\right]$$

with error at most $t^2 \omega_\rho(|t|/\sqrt{m+1})$, uniformly in n . Letting first $n \rightarrow \infty$ and then $m \rightarrow \infty$ gives

$$e^{-(1-\tau_a)t^2/2} \prod_j \phi(a_j t). \quad (52)$$

Thus the limit is $\sum_j a_j X_j$ plus an independent $N(0, 1 - \tau_a)$ variable.

Under the convergence hypothesis this limiting law is the law of X . Suppose that no $|a_j|$ is one. If all a_j vanish, (52) already makes X Gaussian. Otherwise $m_a := \max_j |a_j|$ exists and lies in $(0, 1)$. For $\mathbf{j} = (j_1, \dots, j_k) \in \mathbb{N}^k$, set $b_{\mathbf{j}} := a_{j_1} \cdots a_{j_k}$. Iterating the characteristic-function identity (52) k times gives

$$\phi(t) = e^{-(1-\tau_a^k)t^2/2} \prod_{\mathbf{j} \in \mathbb{N}^k} \phi(t b_{\mathbf{j}}), \quad (53)$$

where

$$\sum_{\mathbf{j}} b_{\mathbf{j}}^2 = \tau_a^k, \quad \sup_{\mathbf{j}} |b_{\mathbf{j}}| \leq m_a^k.$$

The substitution is legitimate in L^2 at every finite depth. Applying lemma 3.7 to (53) yields

$$|\phi(t) - e^{-t^2/2}| \leq t^2 \tau_a^k \omega_\rho(|t| m_a^k) \rightarrow 0.$$

This contradicts non-Gaussianity. Hence some $a_j = \varsigma \in \{-1, 1\}$. All other coefficients and the Gaussian summand vanish, and the limiting identity gives $\text{Law}(\varsigma X) = \rho$.

If (51) failed, a subsequence would stay a fixed distance from every allowed signed unit vector. The preceding extraction produces an allowed ςe_j with

$$\|a^{(n)} - \varsigma e_j\|_2^2 = 2 - 2\varsigma a_j^{(n)} \rightarrow 0.$$

Undoing the permutations is a contradiction. $\square$

Lemma 3.9 (Uniform invariance under tail symmetries). *Let μ be the i.i.d. product law and let $\nu \ll \mu$. Let $\mathcal{T}_N$ consist of the members of $\mathcal{P}$ that change only finitely many coordinates and fix coordinates $1, \dots, N$. Then*

$$\sup_{S \in \mathcal{T}_N} h(\nu, S_{\#} \nu) \rightarrow 0. \quad (54)$$

Also, the one-coordinate marginals of ν converge to the common coordinate law of μ , uniformly over coordinates greater than N .

Proof. Put $\ell = d\nu/d\mu$ and $\ell_N = \mathbb{E}_\mu[\ell \mid \sigma(x_1, \dots, x_N)]$. The martingale convergence theorem gives $\|\ell - \ell_N\|_1 \rightarrow 0$. Every $S \in \mathcal{T}_N$ preserves μ and satisfies $\ell_N \circ S^{-1} = \ell_N$. Thus

$$h(\nu, S_{\#} \nu)^2 \leq \|\ell - \ell \circ S^{-1}\|_{L^1(\mu)} \leq 2\|\ell - \ell_N\|_{L^1(\mu)},$$

which proves (54). For $i > N$, the marginal density is $\mathbb{E}_\mu[\ell \mid x_i]$, while independence gives $\mathbb{E}_\mu[\ell_N \mid x_i] = 1$, so

$$\|\mathbb{E}_\mu[\ell \mid x_i] - 1\|_{L^1(\mu)} \leq \|\ell - \ell_N\|_{L^1(\mu)}.$$

Taking the supremum over $i > N$ proves the marginal assertion. $\square$

Lemma 3.10 (Compactness of the covariance defect). *Let T be bounded on ℓ^2 . If $\nu_T \ll \mu$, then*

$$TT^* - I \text{ is compact.}$$

*If T is boundedly invertible, $T^*T - I$ is compact as well.*

Proof. We first show that, for every $B < \infty$, the squares of

$$Y_b := \sum_j b_j X_j, \quad \|b\|_2 \leq B, \quad (55)$$

are uniformly integrable. Let

$$X^{(M)} = X \mathbf{1}_{\{|X| \leq M\}} - \mathbb{E}[X \mathbf{1}_{\{|X| \leq M\}}], \quad Y_b^{(M)} = \sum_j b_j X_j^{(M)}.$$

The variables $X^{(M)}$ are bounded and centered, and converge to X in L^2 . Independence gives

$$\sup_{\|b\|_2 \leq B} \|Y_b - Y_b^{(M)}\|_2 \leq B \|X - X^{(M)}\|_2 \longrightarrow 0.$$

For fixed M , the fourth-moment formula for independent centered variables gives

$$\sup_{\|b\|_2 \leq B} \mathbb{E}|Y_b^{(M)}|^4 < \infty.$$

It also makes the finite partial sums Cauchy in L^4 , so the bound holds for countable sums. Moreover,

$$\sup_{\|b\|_2 \leq B} \|Y_b^2 - (Y_b^{(M)})^2\|_1 \leq B^2 \|X - X^{(M)}\|_2 (\|X\|_2 + \|X^{(M)}\|_2) \longrightarrow 0.$$

Thus Y_b^2 is uniformly approximated in L^1 by a uniformly integrable family, proving the claim using only the second moment of X .

Write $\nu = \nu_T$, $\ell = d\nu/d\mu$, and $\ell_N = \mathbb{E}_\mu[\ell \mid \sigma(x_1, \dots, x_N)]$. For a finitely supported unit vector a supported on coordinates greater than N , put $F_a(x) = (\sum_i a_i x_i)^2$. Under μ , these are squares from (55); under ν , they are the squares of Y_{T^*a} under μ . Since $\|T^*a\|_2 \leq \|T\|_{\text{op}}$, they are uniformly integrable under both laws. The bounded statistic $F_a \wedge L$ depends only on tail coordinates, so independence gives

$$|\mathbb{E}_\nu(F_a \wedge L) - \mathbb{E}_\mu(F_a \wedge L)| \leq L \|\ell - \ell_N\|_1.$$

Letting first $N \rightarrow \infty$ and then $L \rightarrow \infty$, uniformly in a , gives

$$\sup_{\substack{\|a\|_2=1, \text{ } a \text{ finitely supported} \\ \text{supp } a \subset \{N+1, N+2, \dots\}}} |\langle (TT^* - I)a, a \rangle| \longrightarrow 0.$$

Since $TT^* - I$ is self-adjoint, density of finite vectors in the tail subspace implies

$$\|(I - P_N)(TT^* - I)(I - P_N)\|_{\text{op}} \longrightarrow 0.$$

The complementary rows and columns are finite rank, so $TT^* - I$ is compact. If T is invertible, then $T^*T - I = T^{-1}(TT^* - I)T$. $\square$

We now pass from qualitative row matching to square-summable error. Compactness of the covariance defect normalizes the remote rows; tail marginal convergence and [lemma 3.8](#) match each with an allowed signed coordinate; bounded invertibility makes these matches eventually

distinct; and $T^*T - I$ gives the corresponding column matching. None of these facts controls the sum of the errors.

Suppose that sum diverges. Isolate a remote finite packet with fixed small error energy and swap it with fresh, almost perfectly matched auxiliary coordinates. Conjugating the output swap through T and cancelling the aligned input swap produces $W = PT^{-1}ST$: a small Hilbert–Schmidt perturbation of the identity whose squared norm is comparable to the packet’s error energy. Local coercivity must detect W , while absolute continuity makes the same change asymptotically invisible because S is supported arbitrarily far in the tail. The block computation below has one purpose: to verify this norm comparison without losing constants to the ambient dimension.

Lemma 3.11 (Hilbert–Schmidt necessity under a second-moment hypothesis). *Assume that the common law has an a.e.-positive density f , that $g = \sqrt{f} \in W^{1,2}(\mathbb{R})$, and that $xg' \in L^2(\mathbb{R})$. If $T \in \text{GL}(H)$ and $\nu_T \sim \mu$, then*

$$T - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}. \quad (56)$$

When T is orthogonal, the condition $xg' \in L^2$ can be omitted.

Proof. By lemma 3.10, $\|T^*e_i\|_2 \rightarrow 1$, and by lemma 3.9, the law of the corresponding row sum tends to ρ . Normalization changes that sum by $o_{L^2}(1)$, so lemma 3.8 gives indices $\iota(i)$ and ρ -preserving signs ς_i such that

$$\widehat{e}_i := \varsigma_i e_{\iota(i)}, \quad \|T^*e_i - \widehat{e}_i\|_2^2 \rightarrow 0. \quad (57)$$

The indices $\iota(i)$ are eventually distinct: if $\iota(i) = \iota(j)$, then $\varsigma_i T^*e_i - \varsigma_j T^*e_j$ has arbitrarily small norm by (57), whereas bounded invertibility gives the lower bound $\sqrt{2}/\|T^{-1}\|_{\text{op}}$. Fix N_0 so that ι is injective on $i > N_0$.

Since $\iota(i)$ leaves every finite set, compactness of $T^*T - I$ gives

$$\|T\widehat{e}_i\|_2^2 \rightarrow 1.$$

Together with $\langle T\widehat{e}_i, e_i \rangle = \langle \widehat{e}_i, T^*e_i \rangle \rightarrow 1$, this yields

$$\text{err}_i := \|T^*e_i - \widehat{e}_i\|_2^2 + \|T\widehat{e}_i - e_i\|_2^2 \rightarrow 0. \quad (58)$$

We claim that $\sum_{i>N_0} \text{err}_i < \infty$. Choose the local constants in lemma 3.6 so that

$$h(\mu, \nu_W) \geq c_{\text{HS}} \|W - I\|_{\mathcal{S}_2} \quad \text{when } W \in \text{GL}(H), \quad \|W - I\|_{\mathcal{S}_2} \leq \delta_{\text{HS}}. \quad (59)$$

Suppose the claim fails, and choose $\delta > 0$ so that $3\|T^{-1}\|_{\text{op}}\delta < \delta_{\text{HS}}$. In an arbitrarily remote tail where $\text{err}_i \leq \delta^2$, choose a finite set E satisfying

$$\delta^2 \leq \sum_{i \in E} \text{err}_i \leq 2\delta^2. \quad (60)$$

Pair each $i \in E$ with a distinct later index k_i such that, for $G = \{k_i : i \in E\}$, the sum $\sum_{k \in G} \text{err}_k$ is arbitrarily small.

Let S exchange each output pair (e_i, e_{k_i}) , and let P exchange the aligned signed input pairs $(\widehat{e}_i, \widehat{e}_{k_i})$, acting identically on their orthogonal complements. Both are involutions in $\mathcal{P}$, so they preserve μ . Define

$$W = PT^{-1}ST.$$

Then

$$W - I = PT^{-1}(ST - TP), \quad \frac{\|ST - TP\|_{\mathcal{S}_2}}{\|T\|_{\text{op}}} \leq \|W - I\|_{\mathcal{S}_2} \leq \|T^{-1}\|_{\text{op}} \|ST - TP\|_{\mathcal{S}_2}. \quad (61)$$

Hellinger invariance and $P_{\#}\mu = \mu$ give

$$h(\mu, \nu_W) = h(\nu_T, S_{\#}\nu_T). \quad (62)$$

All differences have finite rank, and [lemma 3.1](#) supplies the common carrier needed for this identity.

To compare the norms, freeze the auxiliary matches by replacing in T each row and column indexed by G with its matched coordinate vector:

$$T_0^* e_k = \widehat{e}_k, \quad T_0 \widehat{e}_k = e_k \quad (k \in G).$$

and retain the complementary block of T . Orthogonal block decomposition gives

$$\|T - T_0\|_{\mathcal{S}_2}^2 \leq \sum_{k \in G} (\|T^* e_k - \widehat{e}_k\|_2^2 + \|T \widehat{e}_k - e_k\|_2^2) = \sum_{k \in G} \text{err}_k. \quad (63)$$

Using the decompositions $E \oplus (E \cup G)^c \oplus G$ on the output and the corresponding aligned input coordinates, write

$$T_0 = \begin{pmatrix} A & B & 0 \\ C & D & 0 \\ 0 & 0 & I \end{pmatrix}, \quad ST_0 - T_0 P = \begin{pmatrix} 0 & -B & I - A \\ C & 0 & -C \\ A - I & B & 0 \end{pmatrix}.$$

Here B, C are Hilbert–Schmidt, and the error on E is

$$\sum_{i \in E} (\|T_0^* e_i - \widehat{e}_i\|_2^2 + \|T_0 \widehat{e}_i - e_i\|_2^2) = 2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2.$$

while the displayed commutator gives

$$\|ST_0 - T_0 P\|_{\mathcal{S}_2}^2 = 2(\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2),$$

Therefore

$$\begin{aligned} 2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2 &\leq \|ST_0 - T_0 P\|_{\mathcal{S}_2}^2 \\ &\leq 2(2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2). \end{aligned} \quad (64)$$

As $\sum_{k \in G} \text{err}_k \rightarrow 0$, the first displayed error tends to $\sum_{i \in E} \text{err}_i$, while

$$\|(ST - TP) - (ST_0 - T_0 P)\|_{\mathcal{S}_2} \leq 2\|T - T_0\|_{\mathcal{S}_2} \rightarrow 0.$$

Hence (60) and (64) allow the auxiliary indices to be chosen so that

$$\frac{\delta}{2} \leq \|ST - TP\|_{\mathcal{S}_2} \leq 3\delta.$$

It follows from (61) that

$$\frac{\delta}{2\|T\|_{\text{op}}} \leq \|W - I\|_{\mathcal{S}_2} \leq 3\|T^{-1}\|_{\text{op}}\delta < \delta_{\text{HS}}.$$

By (59), $h(\mu, \nu_W) \geq c_{\text{HS}}\delta/(2\|T\|_{\text{op}})$. But S is supported in the chosen remote tail, so lemma 3.9 and (62) make this quantity tend to zero as the tail recedes. This contradiction proves $\sum_{i>N_0} \text{err}_i < \infty$.

Define Q_0 by the matched rows $\hat{e}_i$ after the fixed initial cutoff N_0 , and by zero rows for $i \leq N_0$. The summability just proved implies

$$\begin{aligned} \|T - Q_0\|_{\mathcal{S}_2}^2 &= \sum_{i \leq N_0} \|T^* e_i\|_2^2 + \sum_{i > N_0} \|T^* e_i - \hat{e}_i\|_2^2 \\ &\leq \sum_{i \leq N_0} \|T^* e_i\|_2^2 + \sum_{i > N_0} \text{err}_i < \infty. \end{aligned}$$

Thus Q_0 is a compact perturbation of the invertible operator T , hence Fredholm of index zero. Its cokernel is the span of the first N_0 output coordinates and its kernel is spanned by the unused input coordinates, so exactly N_0 columns are unused. Matching them to the first N_0 rows with positive signs completes Q_0 to $Q \in \mathcal{P}$ and proves (56).

If T is orthogonal, both covariance defects vanish, and every $W = PT^*ST$ above is orthogonal. Thus only the orthogonal part of lemma 3.6 is used, and it requires no scale Fisher information. $\square$

3.6 Exact identification of the linear stabilizer

Unlike measure-class equivalence, exact invariance requires only the second moment.

Proof of corollary 1.2. Every member of $\mathcal{P}$ preserves the product law. Conversely, if $\nu_T = \mu$, covariance gives $TT^* = I$. Since T is boundedly invertible, it is orthogonal. Each of its unit rows gives a linear combination with the same law as X . Applying lemma 3.8 to the constant sequence consisting of this row shows that the row is an allowed signed unit vector. Orthogonality makes the selected coordinates a bijection. Thus $T \in \mathcal{P}$. $\square$

3.7 Two-sided uniform integrability and completion of the proof

The finite-coordinate limits and the two finite-marginal conditional-expectation martingales now combine to complete the proof.

Proposition 3.12 (Finite-coordinate approximants recover the likelihood ratio). *Under the assumptions of theorem 1.1, suppose $\nu_T \sim \mu$. For any $Q \in \mathcal{P}$ satisfying $TQ^{-1} - I \in \mathcal{S}_2(H)$, put $K = TQ^{-1} - I$ and $T_n = I + P_n K P_n$. Then T_n is eventually invertible and*

$$\frac{d\nu_{T_n}}{d\mu} \longrightarrow \frac{d\nu_{TQ^{-1}}}{d\mu} \quad \text{in } L^1(\mu), \quad \sqrt{\frac{d\nu_{T_n}}{d\mu}} \longrightarrow \sqrt{\frac{d\nu_{TQ^{-1}}}{d\mu}} \quad \text{in } L^2(\mu). \quad (65)$$

Since $\nu_Q = \mu$, the limiting law is also ν_T . Moreover, the finite-marginal likelihood ratios and their reciprocals are uniformly integrable in their respective reference measures.

Proof of theorem 1.1 and proposition 3.12. Necessity follows from lemma 3.11. If $T - Q \in \mathcal{S}_2$ for $Q \in \mathcal{P}$, then $TQ^{-1} = I + (T - Q)Q^{-1}$ is invertible and differs from the identity by a Hilbert–Schmidt operator. By proposition 3.4, $\nu_{TQ^{-1}} \sim \mu$. Since $Q_{\#}\mu = \mu$, this is $\nu_T \sim \mu$. The same proposition gives the finite-coordinate likelihood-ratio and square-root limits stated in proposition 3.12. The exact linear stabilizer is identified in corollary 1.2. Finally, lemmas 3.2 and 3.3 give the finite-marginal affinity characterization and the singular alternative in theorem 1.1, and both uniform-integrability statements in proposition 3.12. $\square$

Proof of corollary 1.3. Necessity is lemma 3.11. For sufficiency, write $UQ^{-1} = I + (U - Q)Q^{-1}$. This is orthogonal and differs from the identity by a Hilbert–Schmidt operator. Apply proposition 3.5, and then use $Q_{\#}\mu = \mu$. The alternative follows from lemma 3.2. $\square$

3.8 Globalization of finite-dimensional coercivity

Local coercivity controls only a fixed Frobenius neighborhood of the stabilizer. The global obstacle would be a sequence in varying dimensions whose Hellinger displacement tends to zero while its distance from every allowed signed permutation stays positive. After passing to a subsequence whose block Hellinger energies are summable, we stack these matrices as blocks of one bounded invertible operator on ℓ^2 . Kakutani’s theorem turns this summability into equivalence of the infinite product, and the classification forces the block-diagonal operator to be Hilbert–Schmidt-close to one allowed signed permutation. That permutation must eventually preserve each block, contradicting the assumed separation. This direct-sum argument is the compactness input missing from the local theorem.

Proof of theorem 2.7. Fix $0 < a \leq b < \infty$. We first prove uniform separation away from the exact linear stabilizer: for every $\varepsilon > 0$,

$$\inf \left\{ h(\mu_n, A_{\#}\mu_n)^2 : \begin{array}{l} n \geq 1, A \in \text{GL}(n, \mathbb{R}), \\ s_{\min}(A) \geq a, s_{\max}(A) \leq b, \\ d_n(A) \geq \varepsilon \end{array} \right\} > 0. \quad (66)$$

If this fails, choose matrices $A_k \in \text{GL}(n_k, \mathbb{R})$ in the displayed class whose Hellinger distances tend to zero. Since

$$-2 \log A(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) = -2 \log \left(1 - \frac{1}{2} h(\mu_{n_k}, (A_k)_{\#}\mu_{n_k})^2 \right),$$

we may pass to a subsequence for which the sum of these block energies is finite.

On consecutive coordinate blocks define $A = \bigoplus_{k \geq 1} A_k \in \text{GL}(H)$. Each block law is equivalent to μ_{n_k} , and the energy summability gives

$$\prod_{k \geq 1} A(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) > 0.$$

Kakutani’s theorem [17] gives $\nu_A \sim \mu$, so theorem 1.1 supplies $Q \in \mathcal{P}$ with

$$D := A - Q \in \mathcal{S}_2(H). \quad (67)$$

Let Π_k be the orthogonal projection onto the k -th output block. The orthogonality of these blocks gives

$$\sum_k \|\Pi_k D\|_{\mathcal{S}_2}^2 = \|D\|_{\mathcal{S}_2}^2 < \infty, \quad \text{hence} \quad \|\Pi_k D\|_{\mathcal{S}_2} \longrightarrow 0.$$

If e_i belongs to the k th output block but Q^*e_i lies outside the corresponding input block, then the two vectors have disjoint supports and

$$\|A^*e_i - Q^*e_i\|_2^2 = \|A_k^*e_i\|_2^2 + 1 \geq a^2 + 1. \quad (68)$$

For large k this contradicts $\|\Pi_k D\|_{\mathcal{S}_2}^2 < a^2 + 1$. Thus Q^* preserves every sufficiently large block; its restriction is a bijection and defines $Q_k \in \mathcal{P}_{n_k}$. Hence

$$d_{n_k}(A_k) \leq \|A_k - Q_k\|_F = \|\Pi_k D\|_{\mathcal{S}_2} \longrightarrow 0,$$

contrary to $d_{n_k}(A_k) \geq \varepsilon$. This proves (66).

Let $c_{\text{loc}}, C_{\text{loc}}, \delta_{\text{loc}}$ be the local constants, with $0 < \delta_{\text{loc}} < 1$. Choose a minimizer $Q_n \in \mathcal{P}_n$ for $d_n(T)$. Since $(Q_n)_{\#}\mu_n = \mu_n$,

$$h(\mu_n, T_{\#}\mu_n) = h(\mu_n, (TQ_n^{-1})_{\#}\mu_n), \quad \|TQ_n^{-1} - I\|_F = d_n(T).$$

For $d_n(T) \leq \delta_{\text{loc}}$, the local theorem gives the required bounds. For $d_n(T) > \delta_{\text{loc}}$, (66) gives the lower bound and $h^2 \leq 2$ the upper bound. Taking c no larger than c_{loc}^2 and the separation constant at $\varepsilon = \delta_{\text{loc}}$, and taking

$$C = \max\{C_{\text{loc}}^2, 2\delta_{\text{loc}}^{-2}\}.$$

proves (34). $\square$

4 Symmetric Stable Products

The finite-variance criterion does not extend unchanged to symmetric stable products: both the coefficient domain and the mixing cost become α -dependent. For product-preserving affine transformations of independent stable coordinates, equivalence and singularity criteria were obtained by Marques and Cambanis [23]. For Cauchy products, Lie and Sullivan characterize equivalence under coordinatewise dilations [20], and Okamura allows both location and scale parameters to vary [26]. The column cost below addresses the dependence created by off-diagonal mixing. Fix $0 < \alpha < 2$, let ρ_α be the standard symmetric α -stable law, and put

$$\int_{\mathbb{R}} e^{itx} \rho_\alpha(dx) = e^{-|t|^\alpha}, \quad \mu_\alpha = \rho_\alpha^{\otimes \mathbb{N}}, \quad \mu_{\alpha,n} = \rho_\alpha^{\otimes n}. \quad (69)$$

Let $(X_j)_{j \geq 1}$ be independent with common law ρ_α , and write $\mathbf{X} = (X_j)_{j \geq 1}$. We use the symmetric, unit-scale characteristic-function convention; stable parameterizations and their location conventions are described in [25]. Its density f_α is smooth and strictly positive, and its location and scale Fisher information are finite, although its second moment is infinite [4]. Indeed, the density and derivative asymptotics in [40] give $f_\alpha(x) \asymp (1 + |x|)^{-1-\alpha}$ and its location score is $O((1 + |x|)^{-1})$ in the tails, so both the location score and its product with x are square integrable against f_α . For $\alpha \leq 1$, the normalization in (69) specifies a zero location parameter, not a finite mean.

A stable row series $\sum_j a_j X_j$ converges almost surely precisely when $a \in \ell^\alpha$; see the stable-series theory in [30]. For a real matrix T with ℓ^α rows, write $\Phi_{\alpha,T}$ for the associated row-series map and $\nu_{\alpha,T} = (\Phi_{\alpha,T})_{\#}\mu_\alpha$. We say that T admits a compatible stable row-series realization if there is a matrix S with ℓ^α rows such that the two row-series maps are measurable and

$$\Phi_{\alpha,S} \circ \Phi_{\alpha,T} = I, \quad \Phi_{\alpha,T} \circ \Phi_{\alpha,S} = I$$

on a common measurable carrier of full measure for the corresponding orbit laws. This is the stable analogue of the compatible realization supplied by lemma 3.1; it is an assumption on the row-series realization, not a claim that every bounded ℓ^2 operator has stable row sums.

For a matrix K , define its column cost by

$$F_\alpha(K) = \sum_j \|K e_j\|_2^\alpha.$$

For $M = (m_{ij})$, define the anisotropic mixing cost

$$\mathfrak{e}_\alpha(M) = \sum_i |m_{ii} - 1|^2 + F_\alpha(M - \text{diag } M). \quad (70)$$

The first term measures coordinate scales. The second takes the ℓ^2 -norm down each off-diagonal input column and then its α -th power across source coordinates. For the actual stable output marginals, set

$$E_{\alpha,n}(T) = -2 \log A(\mu_{\alpha,n}, (\pi_n)_\# \nu_{\alpha,T}). \quad (71)$$

Theorem 4.1 (Complete criterion for symmetric stable coordinates). *Let $0 < \alpha < 2$ and suppose that T admits a compatible stable row-series realization. Then*

$$\begin{aligned} \nu_{\alpha,T} \sim \mu_\alpha &\iff T|_H \in \text{GL}(H) \text{ and } \mathfrak{e}_\alpha(TQ^{-1}) < \infty \text{ for some } Q \in \mathcal{P}^\pm \\ &\iff \sup_n E_{\alpha,n}(T) < \infty, \end{aligned}$$

(72)

Here $\mathfrak{e}_\alpha$ is the anisotropic mixing cost from (70), whose off-diagonal component is the column cost F_α . If these conditions fail, then $\nu_{\alpha,T} \perp \mu_\alpha$ and $E_{\alpha,n}(T) \uparrow \infty$. Conversely, if $T \in \text{GL}(H)$ satisfies the summability condition in (72), that condition itself supplies the compatible stable row-series realization for T and its inverse. The exact linear stabilizer in this class is $\mathcal{P}^\pm$.

Proof. The proof has three parts: the translation action gives the dichotomy, a Gaussian scale mixture proves sufficiency, and necessity follows from a Fourier transfer to the finite-variance theorem followed by a rare-event test that recovers the sharper column cost.

Dichotomy, energy, and the coefficient domain. Put $g_\alpha = \sqrt{f_\alpha}$ and define

$$a_\alpha(u) = \int_{\mathbb{R}} g_\alpha(x) g_\alpha(x - u) dx.$$

Finite positive location Fisher information and decay of translations in L^2 give

$$1 - a_\alpha(u) \asymp_\alpha \min\{u^2, 1\}. \quad (73)$$

Kakutani's theorem therefore identifies ℓ^2 as the translation space of μ_α . Under equivalence, conjugating translations through the compatible realization shows that T and its inverse matrix S preserve ℓ^2 ; the closed graph theorem gives $T|_H \in \text{GL}(H)$.

The two-sided inverse also makes every finite family of output rows linearly independent. A nonsingular input minor, convolved with the remaining stable inputs, gives each finite output block an everywhere-positive density, so its likelihood ratios form the martingale of [lemma 3.3](#). Both product measures are quasi-invariant and ergodic under their corresponding finitely supported translation groups; the equivalent-singular theorem therefore applies [27]. It follows exactly as in (39) that

$$\nu_{\alpha,T} \sim \mu_\alpha \iff \sup_n E_{\alpha,n}(T) < \infty, \quad (74)$$

with singularity and divergent energy on the other branch.

Sufficiency. The column cost satisfies

$$F_\alpha(AK) \leq \|A\|_{\text{op}}^\alpha F_\alpha(K), \quad F_\alpha(KD) \leq \|D\|_{\text{op}}^\alpha F_\alpha(K) \quad (75)$$

for bounded A and bounded diagonal D , and

$$\|K^* \gamma\|_\alpha^\alpha \leq F_\alpha(K) \|\gamma\|_2^\alpha. \quad (76)$$

Thus $F_\alpha(K) < \infty$ gives admissible stable rows.

The finite-dimensional estimate that replaces the finite-variance Hellinger path bound is

$$h^2(A_{\#} \mu_{\alpha,n}, \mu_{\alpha,n}) \leq C_\alpha \min\{\mathfrak{e}_\alpha(A), 1\} \quad (77)$$

for every invertible real $n \times n$ matrix A , with a constant independent of n . Put $\beta = \alpha/2$, and let V_i be independent positive β -stable variables with $\mathbb{E}e^{-tV_i} = e^{-t^\beta}$, and write $X_i = \sqrt{2V_i}Z_i$ with independent standard Gaussians Z_i . This is the positive-stable subordination representation [30, 31]. Conditioning on $V = \text{diag}(V_i)$, Hellinger data processing and the Gaussian affinity formula bound the left side of (77) by a constant times

$$\mathbb{E} \min\{\|V^{-1/2}(A - I)V^{1/2}\|_F^2, 1\}.$$

For $c_j^2 = \sum_{i \neq j} |a_{ij}|^2$, the contribution of source column j is

$$W_j = V_j \sum_{i \neq j} \frac{|a_{ij}|^2}{V_i}.$$

The positive-stable estimates

$$\mathbb{P}(V_1 > y) \leq C_\beta y^{-\beta}, \quad \mathbb{E}[V_1; V_1 \leq y] \leq C_\beta y^{1-\beta}, \quad \mathbb{E}V_1^{-1} < \infty$$

bound the contribution of $\{c_j^2 V_j > 1\}$ by $C_\alpha c_j^{2\beta} = C_\alpha c_j^\alpha$, while on the complementary event independence gives

$$\mathbb{E}[W_j; c_j^2 V_j \leq 1] \leq C_\beta c_j^{2\beta} \mathbb{E}[V_j; V_j \leq c_j^{-2}] \leq C_\alpha c_j^\alpha.$$

Thus column j costs at most $C_\alpha c_j^\alpha$. Summing over j and using the ordinary quadratic estimate for diagonal scales proves (77).

Remove the measure-preserving Q and write $M = D + K$, where D is diagonal with entries d_i , $(d_i - 1) \in \ell^2$, and $F_\alpha(K) < \infty$. Absorb into K the finitely many nonpositive diagonal entries of M . For

$$D_n = I + P_n(D - I)P_n, \quad K_n = P_n K P_n, \quad M_n = D_n + K_n,$$

Then $M_n \rightarrow M$ in operator norm, so they are eventually invertible with uniformly bounded inverses. From

$$M_n^{-1} = D_n^{-1} - M_n^{-1} K_n D_n^{-1}$$

and (75), $F_\alpha(M_n^{-1} K_n D_n^{-1})$ is uniformly bounded. Also

$$\begin{aligned} M_n^{-1} M_m - I &= D_n^{-1} (D_m - D_n) - M_n^{-1} K_n D_n^{-1} (D_m - D_n) \\ &\quad + M_n^{-1} (K_m - K_n). \end{aligned} \quad (78)$$

The first term tends to zero in diagonal ℓ^2 ; the other two tend to zero in column cost by (75), $\|D_m - D_n\|_{\text{op}} \rightarrow 0$, and $F_\alpha(K_m - K_n) \rightarrow 0$. Their diagonal square costs also vanish because finite column cost implies Hilbert–Schmidt membership. Hence (77) makes the square-root likelihoods of $(M_n)_{\#} \mu_\alpha$ Cauchy in L^2 . The limit has norm one, and ℓ^α -convergence of every row

identifies its square with $d\nu_{\alpha,M}/d\mu_\alpha$. This proves absolute continuity, which (74) upgrades to equivalence. Finally,

$$(D + K)^{-1} = D^{-1} - (D + K)^{-1}KD^{-1} \quad (79)$$

has the same admissible form by (75). Continuity of the stable random functional from ℓ^α to convergence in probability passes the finite-matrix composition identities to the row-series limits, giving compatible measurable versions.

Reduction to a Hilbert–Schmidt perturbation. Conversely, suppose $\nu_{\alpha,T} \sim \mu_\alpha$. We first transfer the half-density through Fourier transform to obtain the Hilbert–Schmidt part; the sharper column cost will be recovered separately.

Let $\hat{g}_\alpha$ be the unitary Fourier transform of g_α , and set

$$\lambda(d\xi) = |\hat{g}_\alpha(\xi)|^2 d\xi.$$

The law λ is symmetric, has variance $\|g'_\alpha\|_2^2 \in (0, \infty)$, and has characteristic function a_α . Since $\hat{g}_\alpha * \hat{g}_\alpha$ is a positive multiple of $e^{-|t|^\alpha}$, we also have $|\hat{g}_\alpha|^2 * |\hat{g}_\alpha|^2 > 0$ almost everywhere. Polynomial stable tails give, for some $m_0, \delta_0 > 0$,

$$c(1 + |t|)^{-m_0} \leq a_\alpha(t) \leq C(1 + |t|)^{-\delta_0},$$

and a'_α is bounded. Choose an integer r so large that

$$\int_{\mathbb{R}} t^2 (|a_\alpha(t)|^{2r} + |(a_\alpha^r)'(t)|^2) dt < \infty.$$

Fourier inversion gives λ^{*r} a bounded-variation density; its third convolution has finite location Fisher information by the Fisher smoothing estimate [4]. Let f_λ be the density of λ^{*3r} , put $f_{\lambda,2} = f_\lambda * f_\lambda$, and let $s_\lambda, s_{\lambda,2}$ be their location scores. Independent U_1, U_2 with density f_λ satisfy

$$xs_{\lambda,2}(x) - 1 = 2\mathbb{E}[U_1 s_\lambda(U_2) \mid U_1 + U_2 = x].$$

Finite variance and finite Fisher information of f_λ therefore give finite scale Fisher information for $f_{\lambda,2}$. Set $k = 6r$ and $\Lambda_k = (\lambda^{*k})^{\otimes \mathbb{N}}$. After rescaling to variance one, λ^{*k} has an a.e.-positive density satisfying (1); it is non-Gaussian because its characteristic function a_α^k has a polynomial lower bound.

Let $f_{\alpha,n} = f_\alpha^{\otimes n}$, let $f_{\alpha,T,n}$ be the density of the true first n stable outputs, and put

$$\ell_{\alpha,T,n} = \frac{f_{\alpha,T,n}}{f_{\alpha,n}}, \quad \ell_{\alpha,T} = \frac{d\nu_{\alpha,T}}{d\mu_\alpha}.$$

Here $\ell_{\alpha,T,n}$ is lifted to the product space and $\sqrt{f_{\alpha,T,n}} = g_\alpha^{\otimes n} \sqrt{\ell_{\alpha,T,n}}$. Equivalence gives $\sqrt{\ell_{\alpha,T,n}} \rightarrow \sqrt{\ell_{\alpha,T}}$ in $L^2(\mu_\alpha)$.

We construct $((T^{-1})^*)_{\#} \Lambda_k$ from finite marginals: Fourier transformation turns half-density translations into phases, tensorization raises autocorrelations to the k th power, and frequency addition realizes convolution by λ^{*k} .

For each n , take k tensor copies of $\sqrt{f_{\alpha,T,n}}$, Fourier transform, square the modulus, and add the k frequency coordinates at each site. Denote the resulting law on $\mathbb{R}^n$ by $\xi_{\alpha,T,n}$, and set

$$\Xi_{\alpha,T,n} = \xi_{\alpha,T,n} \otimes (\lambda^{*k})^{\otimes \{i > n\}}. \quad (80)$$

The first factor has a Lebesgue density and the tail density is positive almost everywhere; hence $\Xi_{\alpha,T,n} \ll \Lambda_k$.

For $m > n$, pad $\sqrt{f_{\alpha,T,n}}$ by $m - n$ factors of g_α . Unitarity, contraction under pushforward, and $\| |u|^2 - |v|^2 \|_1 \leq 2\|u - v\|_2$ give

$$\|\Xi_{\alpha,T,m} - \Xi_{\alpha,T,n}\|_{\text{TV}} \leq 2\|(\sqrt{\ell_{\alpha,T,m}})^{\otimes k} - (\sqrt{\ell_{\alpha,T,n}})^{\otimes k}\|_{L^2(\mu_\alpha^{\otimes k})} \leq 2k\|\sqrt{\ell_{\alpha,T,m}} - \sqrt{\ell_{\alpha,T,n}}\|_{L^2(\mu_\alpha)}. \quad (81)$$

Thus $\Xi_{\alpha,T,n}$ converges in total variation to a probability $\Xi_{\alpha,T} \ll \Lambda_k$.

Let $\nu_{\alpha,T}^{[n]} = \ell_{\alpha,T,n}\mu_\alpha$. For every finitely supported z , Fourier autocorrelation and square-root convergence give

$$\begin{aligned} \widehat{\Xi}_{\alpha,T}(z) &= \lim_n \mathbf{A}(\nu_{\alpha,T}^{[n]}, (x \mapsto x + z)_{\#} \nu_{\alpha,T}^{[n]})^k \\ &= \mathbf{A}(\nu_{\alpha,T}, (x \mapsto x + z)_{\#} \nu_{\alpha,T})^k \\ &= \prod_i a_\alpha((T^{-1}z)_i)^k. \end{aligned}$$

The last expression is the characteristic function of $((T^{-1})^*)_{\#} \Lambda_k$; agreement for finitely supported z identifies all finite marginals. Hence

$$((T^{-1})^*)_{\#} \Lambda_k \ll \Lambda_k. \quad (82)$$

The dichotomy and classification in [theorem 1.1](#) upgrade this to equivalence and give $Q \in \mathcal{P}^\pm$ such that $(T^{-1})^* - Q \in \mathcal{S}_2$. Since

$$T - Q = -T((T^{-1})^* - Q)^*Q,$$

we obtain $T - Q \in \mathcal{S}_2$. After removing Q , write

$$TQ^{-1} = I + K, \quad K \in \mathcal{S}_2. \quad (83)$$

Necessity of the off-diagonal column summability. For $\alpha < 2$, Hilbert–Schmidt control is weaker than the column cost. We detect column j on the event $|X_j| \asymp \|Be_j\|_2^{-1}$, whose probability has the required order. This gives constants $\varepsilon_\alpha, c_\alpha > 0$ such that

$$\text{diag } B = 0, \quad F_\alpha(B) \leq \varepsilon_\alpha \implies h^2((I + B)_{\#} \mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha F_\alpha(B). \quad (84)$$

To prove it, write $F = F_\alpha(B)$, $c_j = \|Be_j\|_2$, and $u_j = Be_j/c_j$ when $c_j \neq 0$. Then $\|u_j\|_2 = 1$, $(u_j)_j = 0$, and the stable scale contributed to output row i satisfies

$$\omega_i(B) := \sum_j |b_{ij}|^\alpha \leq F. \quad (85)$$

For any finite family of Hilbert-space vectors v_j ,

$$\mathbb{P} \left(\left\| \sum_j v_j X_j \right\|_2 > t \right) \leq C_\alpha t^{-\alpha} \sum_j \|v_j\|_2^\alpha. \quad (86)$$

Indeed, exclude $\{\|v_j\|_2|X_j| > t\}$ and apply Chebyshev to the centered truncated sum, using the stable tail and truncated second moment. Integration gives a modulus $\tau_\alpha(s) \downarrow 0$ as $s \downarrow 0$ such that

$$\mathbb{E} \min \left\{ \left\| \sum_j v_j X_j \right\|_2, 1 \right\} \leq \tau_\alpha \left(\sum_j \|v_j\|_2^\alpha \right). \quad (87)$$

Second, one bounded statistic detects a unit displacement in every Euclidean direction, uniformly in the dimension. Take $\psi(x) = \tanh x$. There are a smooth odd increasing function χ , bounded in modulus by one, and $c_{\alpha, \text{rec}} > 0$, depending only on ρ_α , such that for every unit vector u ,

$$\Theta_u(x) = \sum_i u_i \psi(x_i), \quad \mathcal{M}_u(t) = \mathbb{E} \chi(\Theta_u(\mathbf{X} + tu)), \quad \mathcal{M}_u(t) \geq c_{\alpha, \text{rec}} \quad (1 \leq t \leq 2). \quad (88)$$

Oddness and boundedness give $\mathbb{E} \Theta_u(\mathbf{X}) = 0$ and $\text{Var} \Theta_u(\mathbf{X}) = \text{Var}(\psi(X)) \sum_i u_i^2$, so one compact interval contains all but a uniformly small amount of its mass. Moreover,

$$|\Theta_u(\mathbf{X} + tu) - \Theta_u(\mathbf{X})| \leq |t| \sum_i u_i^2 \leq 2.$$

Choose χ so that χ' is positive on that interval enlarged by two. Since

$$\inf_{|\zeta| \leq 2} \mathbb{E} \psi'(X + \zeta) > 0,$$

the expectation of $\sum_i u_i^2 \psi'(X_i + tu_i)$ has the same positive lower bound. Taking the discarded contribution smaller than half of it and differentiating bounds from below

$$\mathcal{M}'_u(t) = \mathbb{E} \left[\chi'(\Theta_u(\mathbf{X} + tu)) \sum_i u_i^2 \psi'(X_i + tu_i) \right]$$

for $|t| \leq 2$. Oddness proves (88).

Now choose a nonzero smooth odd function w , supported where $1 \leq |t| \leq 2$, with $0 \leq w \leq 1$ on the positive half-line. Put

$$\Theta_j(x) = \sum_{i \neq j} (u_j)_i \psi(x_i),$$

and define the bounded test

$$\Psi_B(x) = \max \left\{ -1, \min \left\{ \sum_{j: c_j \neq 0} w(c_j x_j) \chi(\Theta_j(x)), 1 \right\} \right\}. \quad (89)$$

For all sufficiently small c_j , stable tail asymptotics give

$$\mathbb{P}(1 \leq |c_j X_j| \leq 2) \leq C_\alpha c_j^\alpha, \quad \mathbb{E} |w(c_j X_j)| \geq c_\alpha c_j^\alpha. \quad (90)$$

The gates are independent. If N_{open} is their number, then $\mathbb{E}[N_{\text{open}}(N_{\text{open}} - 1)] \leq C_\alpha F^2$. Before truncation, every summand has mean zero because its gate is independent of its receiver variables. Truncation matters only when at least two gates are open, so

$$|\mathbb{E} \Psi_B(\mathbf{X})| \leq C_\alpha F^2. \quad (91)$$

It remains to control the same statistic at $\mathbf{Y} = (I + B)\mathbf{X}$. For fixed $0 < \delta < 1/4$, (85) and the stable tail bound imply

$$\mathbb{P}\{\exists i : |c_i(B\mathbf{X})_i| > \delta\} \leq C_\alpha \delta^{-\alpha} \sum_i c_i^\alpha \omega_i(B) \leq C_\alpha \delta^{-\alpha} F^2. \quad (92)$$

On the complementary event, Lipschitz continuity of w and (90) show that replacing every output gate $w(c_j Y_j)$ by its input gate $w(c_j X_j)$ changes the expectation by at most

$$C_\alpha \delta F + C_\alpha \delta^{-\alpha} F^2. \quad (93)$$

After this replacement, removing the outer truncation costs at most another $C_\alpha F^2$.

Finally, the random vector $B(I - e_j \otimes e_j)\mathbf{X}$ is independent of the gate variable X_j , and (87) shows that replacing $\Theta_j(\mathbf{Y})$ by $\Theta_j(\mathbf{X} + c_j X_j u_j)$ costs, after summing over j , at most

$$C_\alpha F \tau_\alpha(F). \quad (94)$$

Because $(u_j)_j = 0$, conditioning on X_j and using Equations (88) and (90) gives

$$\sum_j \mathbb{E}[w(c_j X_j) \chi(\Theta_j(\mathbf{X} + c_j X_j u_j))] \geq c_\alpha F. \quad (95)$$

Choose δ , then F , so that the δF , F^2 , $\delta^{-\alpha} F^2$, and $F \tau_\alpha(F)$ errors total at most half the main response. Equations (91) and (93) to (95) give

$$\mathbb{E}\Psi_B(\mathbf{Y}) - \mathbb{E}\Psi_B(\mathbf{X}) \geq c_\alpha F.$$

The test is nonzero only when a gate is open, and the coordinates of $\mathbf{Y}$ have stable scales at most $(1 + F)^{1/\alpha}$; hence

$$\mathbb{E}\Psi_B(\mathbf{X})^2 + \mathbb{E}\Psi_B(\mathbf{Y})^2 \leq C_\alpha F.$$

Hellinger Cauchy–Schwarz now gives (84).

We globalize this local estimate using finite tail sign changes. Such a change isolates the entries crossing a coordinate cut: equivalence makes every remote cut Hellinger-invisible, whereas the local estimate detects every cut of a fixed small cost.

Let B have ℓ^α rows, zero diagonal, $\|B\|_{\text{op}} \leq 1/10$, and belong to $\mathcal{S}_2$. Suppose $I + B$ admits a compatible stable row-series realization and $(I + B)_\# \mu_\alpha \sim \mu_\alpha$. For a finite coordinate set $\mathcal{J}$, let $Q_\mathcal{J}$ change the signs in $\mathcal{J}$ and set

$$\Delta_\mathcal{J} = Q_\mathcal{J} B Q_\mathcal{J} - B, \quad A_\mathcal{J} = (I + B)^{-1} Q_\mathcal{J} (I + B) Q_\mathcal{J}.$$

The matrix $\Delta_\mathcal{J}$ consists of the entries crossing the cut. Since $\mathcal{J}$ is finite, it has finite column cost. Applying (84) to $P_n \Delta_\mathcal{J} P_n$ and using the sufficiency approximation extends that bound to $\Delta_\mathcal{J}$.

Put

$$K_\mathcal{J} = (I + B)^{-1} \Delta_\mathcal{J}, \quad D_\mathcal{J} = I + \text{diag } K_\mathcal{J}, \quad \tilde{B}_\mathcal{J} = D_\mathcal{J}^{-1} (I + K_\mathcal{J}) - I.$$

Then $A_\mathcal{J} = I + K_\mathcal{J}$ and $\text{diag } \tilde{B}_\mathcal{J} = 0$. Since $\|(I + B)^{-1} - I\|_{\text{op}} \leq 1/9$, orthogonal projection away from the diagonal entry in each column gives

$$\frac{8}{9} \|\Delta_\mathcal{J} e_j\|_2 \leq \|(I - e_j \otimes e_j) K_\mathcal{J} e_j\|_2 \leq \frac{10}{9} \|\Delta_\mathcal{J} e_j\|_2.$$

For small $F_\alpha(\Delta_{\mathcal{J}})$, the diagonal entries of $D_{\mathcal{J}}$ lie in $[1/2, 3/2]$, and consequently

$$F_\alpha(\tilde{B}_{\mathcal{J}}) \asymp_\alpha F_\alpha(\Delta_{\mathcal{J}}), \quad \|\text{diag } K_{\mathcal{J}}\|_2 \leq \frac{1}{9} F_\alpha(\Delta_{\mathcal{J}})^{1/\alpha}. \quad (96)$$

The lower bound for $\tilde{B}_{\mathcal{J}}$, the diagonal scale upper bound for $D_{\mathcal{J}}$, and the Hellinger triangle inequality now give

$$F_\alpha(\Delta_{\mathcal{J}}) \leq \varepsilon_* \implies h((A_{\mathcal{J}})_{\#} \mu_\alpha, \mu_\alpha) \geq c_* F_\alpha(\Delta_{\mathcal{J}})^{1/2}. \quad (97)$$

Indeed, by (84), the two terms have respective orders $F_\alpha(\Delta_{\mathcal{J}})^{1/2}$ and $F_\alpha(\Delta_{\mathcal{J}})^{1/\alpha}$, and the latter is smaller for small cost because $\alpha < 2$.

Finite-prefix approximation of the square-root likelihood also gives uniform invariance under finite tail sign changes: each fixes the prefix approximation and preserves μ_α , so

$$\sup_{\substack{\mathcal{J} \subset \{N+1, N+2, \dots\} \\ |\mathcal{J}| < \infty}} h((I+B)_{\#} \mu_\alpha, (Q_{\mathcal{J}})_{\#} (I+B)_{\#} \mu_\alpha) \longrightarrow 0. \quad (98)$$

Hellinger invariance and $(Q_{\mathcal{J}})_{\#} \mu_\alpha = \mu_\alpha$ give

$$h((I+B)_{\#} \mu_\alpha, (Q_{\mathcal{J}})_{\#} (I+B)_{\#} \mu_\alpha) = h(\mu_\alpha, (A_{\mathcal{J}})_{\#} \mu_\alpha). \quad (99)$$

Suppose $F_\alpha(B) = \infty$. Deleting finitely many columns or rows changes the cost finitely, the latter because each row lies in ℓ^α ; thus every tail has infinite cost. For a finite tail set I_0 , include each coordinate independently in $\mathcal{J}$ with probability one half. Columnwise concavity of $t^{\alpha/2}$ gives

$$\mathbb{E}_{\mathcal{J}} F_\alpha(P_{I_0} \Delta_{\mathcal{J}} P_{I_0}) \geq 2^{\alpha-1} F_\alpha(P_{I_0} B P_{I_0}). \quad (100)$$

As I_0 exhausts the tail, some finite cuts therefore have arbitrarily large cost. Equivalence makes the remote output marginals converge to ρ_α , so

$$\omega_i(B) := \sum_{j \neq i} |b_{ij}|^\alpha \longrightarrow 0.$$

Since $B \in \mathcal{S}_2$ also gives $c_i := \|B e_i\|_2 \rightarrow 0$, we have

$$F_\alpha(\Delta_{\{i\}}) = 2^\alpha (c_i^\alpha + \omega_i(B)) \longrightarrow 0.$$

For $\alpha \leq 1$, F_α is subadditive; for $\alpha \geq 1$, $F_\alpha^{1/\alpha}$ is a norm. If $i \notin \mathcal{J}$,

$$\Delta_{\mathcal{J} \cup \{i\}} - \Delta_{\mathcal{J}} = Q_{\mathcal{J}} \Delta_{\{i\}} Q_{\mathcal{J}},$$

and signs preserve the relevant subadditive size. Hence single-coordinate increments tend uniformly to zero. Fix $0 < \varepsilon < \varepsilon_*/2$, add the elements of a large cut from (100), and stop at the first threshold crossing. In every sufficiently remote tail this produces $\mathcal{J}$ with $\varepsilon \leq F_\alpha(\Delta_{\mathcal{J}}) \leq 2\varepsilon$. Equations (97) to (99) then contradict one another. Hence $F_\alpha(B) < \infty$.

Finally, write $M = I + K$ as in (83), set $M_n = I + P_n K P_n$, and take n large. Then $A = M_n^{-1} M$ is arbitrarily close to I . Since M_n changes only finitely many coordinates, compatible composition gives $A_{\#} \mu_\alpha \sim \mu_\alpha$. Its diagonal part $D = \text{diag } A$ is positive, boundedly invertible, and $D - I \in \mathcal{S}_2$; the one-coordinate scale criterion therefore gives $D_{\#} \mu_\alpha \sim \mu_\alpha$. Consequently

$$D^{-1} A = I + B, \quad \text{diag } B = 0, \quad B \in \mathcal{S}_2, \quad \|B\|_{\text{op}} \leq 1/10, \quad (I+B)_{\#} \mu_\alpha \sim \mu_\alpha.$$

Finite-coordinate and bounded diagonal composition preserve the ℓ^α row condition and compatible realization, so the finite-cut argument gives $F_\alpha(B) < \infty$. Since K_n is supported on a finite coordinate block, we have

$$M = M_n D(I + B) = D + (K_n D + DB + K_n DB).$$

The parenthesized term has finite column cost by (75). Restoring the signed permutation gives

$$\sum_j \left(\sum_{i \neq j} |k_{ij}|^2 \right)^{\alpha/2} < \infty,$$

while $K \in \mathcal{S}_2$ already gives $\sum_i |k_{ii}|^2 < \infty$. This is $\mathfrak{e}_\alpha(TQ^{-1}) < \infty$.

If $\nu_{\alpha,T} = \mu_\alpha$, comparison of the joint characteristic functions, first for T and then for its compatible inverse, makes T^* a surjective isometry of ℓ^α . For $0 < \alpha < 2$, equality in

$$2\|u\|_\alpha^\alpha + 2\|v\|_\alpha^\alpha - \|u+v\|_\alpha^\alpha - \|u-v\|_\alpha^\alpha \geq 0$$

characterizes disjoint support. Hence the images of the coordinate vectors have disjoint supports; surjectivity makes each support a singleton, and norm preservation makes its entry ± 1 . Thus $T \in \mathcal{P}^\pm$. The converse follows from independence and symmetry. $\square$

4.1 Anisotropic stable Hellinger coercivity and sharpness

The two summands in (70) have genuinely different homogeneities. Let $\mathcal{P}_n^\pm$ denote the group of all $n \times n$ signed permutation matrices. For a finite matrix A , put

$$\mathfrak{e}_{\alpha,n}(A) = \min_{Q \in \mathcal{P}_n^\pm} \mathfrak{e}_\alpha(AQ^{-1}).$$

Proposition 4.2 (Anisotropic stable Hellinger coercivity). *Let $A \in \text{GL}(n, \mathbb{R})$. If all singular values of A lie in a fixed interval $[a, b] \subset (0, \infty)$, then constants depending only on α, a, b , and not on the dimension, satisfy*

$$c_{\alpha,a,b} \min\{\mathfrak{e}_{\alpha,n}(A), 1\} \leq h^2(A \# \mu_{\alpha,n}, \mu_{\alpha,n}) \leq C_{\alpha,a,b} \min\{\mathfrak{e}_{\alpha,n}(A), 1\}. \quad (101)$$

Proof. The upper bound is (77), after removing the minimizing signed permutation. For the even convolution power k used above, Fourier transformation gives

$$h^2(((A^{-1})^*)_{\#}(\lambda^{*k})^{\otimes n}, (\lambda^{*k})^{\otimes n}) \leq k h^2(A \# \mu_{\alpha,n}, \mu_{\alpha,n}). \quad (102)$$

Indeed, Fourier transformation intertwines the half-density actions of A and $(A^{-1})^*$, modulus is an L^2 contraction, tensorization costs at most k , and frequency addition is data processing. Apply finite-variance coercivity to the rescaled λ^{*k} product. The singular values of $(A^{-1})^*$ lie in $[b^{-1}, a^{-1}]$, and

$$A - Q = -A((A^{-1})^* - Q)^* Q$$

compares the two distances up to a factor depending on a, b . Hence

$$h^2(A \# \mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_{\alpha,a,b} \min \left\{ \min_{Q \in \mathcal{P}_n^\pm} \|A - Q\|_F^2, 1 \right\}. \quad (103)$$

Near a signed permutation, remove that permutation and write

$$d_{\text{diag}} = \sum_i |a_{ii} - 1|^2, \quad F = \sum_j \left(\sum_{i \neq j} |a_{ij}|^2 \right)^{\alpha/2}.$$

For $d_{\text{diag}} + F$ small, the identity is the nearest signed permutation, so (103) controls d_{diag} . With $D = \text{diag } A$ and $D^{-1}A = I + B$, $\text{diag } B = 0$ and $F_\alpha(B) \asymp_\alpha F$. The off-diagonal column lower bound and the diagonal scale upper bound give

$$h(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha \sqrt{F} - C_\alpha \sqrt{d_{\text{diag}}}.$$

Choose $\eta > 0$ so that the last negative term is absorbed when $d_{\text{diag}} \leq \eta F$. When $d_{\text{diag}} \geq \eta F$, use (103). This proves the local lower bound.

Otherwise choose a separated sequence with summable squared Hellinger distances and form its block-diagonal sum. Kakutani makes the resulting infinite products equivalent. The classification just proved supplies a signed permutation preserving every sufficiently late block, since crossing one costs a fixed positive square error. Its block restrictions would make the anisotropic costs summable, a contradiction. This proves the global lower bound. $\square$

The different local powers are already visible in two dimensions. For the shear $T_t(x, y) = (x + ty, y)$, conditioning on y and applying Plancherel gives the exact formula

$$h^2((T_t)_{\#}\mu_{\alpha,2}, \mu_{\alpha,2}) = 2 \int_{\mathbb{R}} (1 - e^{-|t\xi|^\alpha}) |\widehat{g}_\alpha(\xi)|^2 d\xi. \quad (104)$$

Consequently

$$h^2((T_t)_{\#}\mu_{\alpha,2}, \mu_{\alpha,2}) \sim 2|t|^\alpha \int_{\mathbb{R}} |\xi|^\alpha |\widehat{g}_\alpha(\xi)|^2 d\xi, \quad t \rightarrow 0,$$

whereas a diagonal scale perturbation has squared Hellinger cost of order t^2 . Thus anisotropic stable Hellinger coercivity is not generated by a quadratic form from one finite-dimensional matrix Fisher operator.

In particular, Hilbert–Schmidt proximity is necessary but no longer sufficient. On independent two-coordinate blocks, let

$$T_a(x_k, y_k)_{k \geq 1} = (x_k + a_k y_k, y_k)_{k \geq 1}.$$

Then $T_a - I \in \mathcal{S}_2$ exactly when $a \in \ell^2$, while theorem 4.1 gives $(T_a)_{\#}\mu_\alpha \sim \mu_\alpha$ exactly when $a \in \ell^\alpha$. Any $a \in \ell^2 \setminus \ell^\alpha$ therefore gives a Hilbert–Schmidt perturbation with singular image law, even though the stable coordinate density is everywhere positive and has finite location and scale Fisher information. Conversely, when one source is mixed into many outputs, the criterion charges the ℓ^2 -norm of that whole column only once. An entrywise ℓ^α condition would therefore be too strong. The correct stable criterion is the anisotropic mixing cost (70), not a Schatten-class replacement for the finite-variance theorem.

5 Density Zeros and Support Rigidity

When the density has zeros, Hilbert–Schmidt proximity controls Hellinger accumulation but not absolute continuity. The missing condition is whether the rows of T and T^{-1} land almost surely in $\{f > 0\}$. We prove this criterion and then show that bounded or one-sided support forces permutation rigidity. Each result states its own hypotheses.

5.1 The metric input when the density has zeros

The Fourier/Fisher mechanism still supplies the metric part of the classification; for its matrix estimates the appropriate regularity is

$$\begin{aligned} f \geq 0, \quad \int f = 1, \quad g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \quad xg' \in L^2(\mathbb{R}), \\ \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad f \text{ non-Gaussian.} \end{aligned} \tag{105}$$

The global zero extension is essential. Membership of g in $W^{1,2}(\mathbb{R})$ rules out a hidden boundary jump and makes its weak derivative vanish almost everywhere on $\{g = 0\}$. Consequently the identity $sg = -2g'$ and the integrations by parts used by the Fourier/Fisher argument remain valid on the whole line, with no boundary term. Interior regularity alone does not suffice: the uniform density has zero interior derivative but a jump in its zero-extended half-density.

Proposition 5.1 (Local Fourier coercivity with zeros of the density). *Under (105), the matrix Fisher operator decomposition, finite-dimensional Hellinger path bound, and dimension-free local Frobenius coercivity in Section 2 remain valid. Write $\mu = \rho^{\otimes \mathbb{N}}$, $\rho(dx) = f(x)dx$. There are constants $c, C, \delta > 0$, with $\delta < 1$, such that*

$$c\|K\|_{\mathcal{S}_2} \leq h(\mu, \nu_{I+K}) \leq C\|K\|_{\mathcal{S}_2} \quad (\|K\|_{\mathcal{S}_2} \leq \delta). \tag{106}$$

Moreover, for every $T \in \text{GL}(H)$, the following necessary condition survives:

$$\nu_T \ll \mu \implies T - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}. \tag{107}$$

For orthogonal transformations one may omit $xg' \in L^2$, retaining non-Gaussianity, $g \in W^{1,2}(\mathbb{R})$, and normalization to a centered law with variance one. These statements do not assert absolute continuity of ν_{I+K} .

Proof. A Sobolev function has weak derivative zero almost everywhere on its zero set. Thus, defining $s = -2g'/g$ on $\{g > 0\}$ and $s = 0$ elsewhere, we have $sg = -2g'$ on $\mathbb{R}$. All cutoff integration-by-parts identities, including $\mathbb{E}s(X) = 0$, $\mathbb{E}[Xs(X)] = 1$, and the formulas for J, κ , remain valid. Moreover, $J = 1$ would give the global Gaussian equation $g' = -xg/2$, while $\kappa = 0$ would give $2xg' + g = 0$, which has no nonzero square-integrable solution. Hence the finite-dimensional Fourier/Fisher proofs apply unchanged.

For $K \in \mathcal{S}_2$ with $I + K$ invertible, put $T_n = I + P_n K P_n$. The Hellinger path bound and invariance give, for large m, n ,

$$h(\nu_{T_n}, \nu_{T_m}) \leq C\|T_n^{-1}\|_{\text{op}}\|T_n - T_m\|_{\mathcal{S}_2}.$$

Relative to one measure dominating $\mu, \nu_{T_1}, \nu_{T_2}, \dots$, the square-root densities are therefore Cauchy in L^2 . Their limit has norm one, and convergence of finite row sums identifies its law with ν_{I+K} . Passing the finite-dimensional bounds to this Hellinger limit proves (106); it does not assert $\nu_{I+K} \ll \mu$. For orthogonal operators, the polar approximants and skew-generator bound from proposition 3.5 give the same conclusion.

The proof of lemma 3.11 uses absolute continuity only through lemmas 3.9 and 3.10, both of which assume only $\nu_T \ll \mu$; its remaining inputs are moment-only rigidity, the measurable inverse action, and the local estimate for finite-rank conjugated tail perturbations. The estimate just proved supplies the last input, so the same tail-matching argument proves (107). $\square$

For a Gamma density centered with variance one, the zero-extended square root is in $W^{1,2}(\mathbb{R})$ exactly when its shape parameter exceeds two. For an affine rescaling of a Beta density, the corresponding condition is that both shape parameters exceed two. These assertions follow by squaring the derivative of the corresponding boundary power. At shape one the square root instead has a jump, so its vanishing interior power derivative is not an exception. The support classifications below require none of this regularity and include every positive Gamma or Beta shape.

5.2 The linear equivalence group for arbitrary zero sets of the density

Hilbert–Schmidt proximity remains the metric condition. The zero-set row condition adds the exact null-set requirement for T and T^{-1} , using the positivity set of the density rather than its closed support.

Theorem 5.2 (The linear equivalence group with density zeros). *Assume (105), let $\rho(dx) = f(x) dx$, and put $\mathcal{U} = \{x \in \mathbb{R} : f(x) > 0\}$. For every $T \in \text{GL}(H)$, the almost-sure requirements below are called the forward and inverse zero-set row conditions.*

$$\nu_T \ll \mu \iff \begin{cases} T - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}, \\ \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i. \end{cases} \quad (108)$$

Consequently the linear equivalence group of μ is

$$\begin{aligned} \mathcal{G}_\rho &:= \{T \in \text{GL}(H) : \nu_T \sim \mu\} \\ &= \left\{ T \in \text{GL}(H) : \begin{cases} T - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}, \\ \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0, \\ \mu \left\{ x : \sum_j (T^{-1})_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i \end{cases} \right\}. \end{aligned} \quad (109)$$

This set is a subgroup of $\text{GL}(H)$. For orthogonal members of $\mathcal{G}_\rho$, the same criterion holds with $T^{-1} = T^*$ and without the scale condition $xg' \in L^2$.

More precisely, whenever $T - Q \in \mathcal{S}_2(H)$ for some $Q \in \mathcal{P}$, the Lebesgue decomposition of ν_T relative to μ is its restriction to $\mathcal{U}^\mathbb{N}$ plus its restriction to the complement: the former is absolutely continuous and the latter is singular.

Proof. Suppose first that $T - Q \in \mathcal{S}_2$ and put $K = TQ^{-1} - I$. Since $\nu_Q = \mu$, we have $\nu_T = \nu_{I+K}$. For the eventually invertible operators $T_n = I + P_n K P_n$, [proposition 5.1](#) gives

$$h(\nu_{T_n}, \nu_T) \longrightarrow 0.$$

Hence $\nu_{T_n} \rightarrow \nu_T$ in total variation. Each ν_{T_n} has a Lebesgue density in its first n coordinates and agrees with μ on the tail; since the density of μ_n is positive on $\mathcal{U}^n$, Fubini gives

$$\nu_{T_n}|_{\mathcal{U}^\mathbb{N}} \ll \mu.$$

Passing to the limit proves $\nu_T|_{\mathcal{U}^{\mathbb{N}}} \ll \mu$, while $\nu_T|_{(\mathcal{U}^{\mathbb{N}})^c} \perp \mu$. Consequently

$$\nu_T \ll \mu \iff \nu_T(\mathcal{U}^{\mathbb{N}}) = 1 \iff \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i.$$

Together with (107), this proves (108).

If $\nu_T \sim \mu$, the measurable inverse action gives $\nu_{T^{-1}} \sim \mu$, so both row conditions are necessary. Conversely, $T - Q \in \mathcal{S}_2$ implies

$$T^{-1} - Q^{-1} = -T^{-1}(T - Q)Q^{-1} \in \mathcal{S}_2.$$

Thus the two row conditions and (108) give $\nu_T, \nu_{T^{-1}} \ll \mu$. The inverse action then yields

$$\mu = (\Phi_T)_{\#} \nu_{T^{-1}} \ll (\Phi_T)_{\#} \mu = \nu_T,$$

and hence equivalence. Finally, if $S, T \in \mathcal{G}_{\rho}$, measurable composition gives $\nu_{ST} = (\Phi_S)_{\#} \nu_T \sim \nu_S \sim \mu$, and the preceding argument gives $T^{-1} \in \mathcal{G}_{\rho}$; thus $\mathcal{G}_{\rho}$ is a group. In the orthogonal case, the same restriction argument uses the approximants from [proposition 3.5](#), which require only location Fisher information. $\square$

Every nonzero row sum has an absolutely continuous scalar law, so the condition is unchanged by modifying f on a null set. It is automatic when $f > 0$ almost everywhere, while the contraction in [Section 5.5](#) satisfies only the forward condition and therefore gives strict absolute continuity rather than equivalence.

5.3 Boundary behavior of the density and summability criteria

The zero-set row condition governs absolute continuity, whereas projected Hellinger energy governs accumulated overlap. Hard and smoothly vanishing boundaries separate these effects.

Let ρ be uniform on $[-\sqrt{3}, \sqrt{3}]$, let $\mu_2 = \rho^{\otimes 2}$, and consider $T = \bigoplus_k O_{\theta_k}$. Write $\delta_k = \min_{m \in \mathbb{Z}} |\theta_k - m\pi/2| \in [0, \pi/4]$. For one block the affinity is the relative area of intersection of a square and its rotation. Removing the four corner triangles gives, for $0 < \delta \leq \pi/4$,

$$\begin{aligned} A(\mu_2, (O_{\delta})_{\#} \mu_2) &= 1 - \frac{(\cos \delta + \sin \delta - 1)^2}{2 \sin \delta \cos \delta}, \\ 1 - A(\mu_2, (O_{\delta})_{\#} \mu_2) &= \frac{\delta}{2} + O(\delta^2). \end{aligned} \tag{110}$$

At $\delta = 0$, the affinity is one. Each nonzero δ leaves positive area on both sides of the support difference. Consequently [theorem 5.4](#) gives the complete trichotomy

$$\begin{array}{l|l} \delta_k = 0 \text{ for every } k & \nu_T = \mu \\ 0 < \sum_k \delta_k < \infty & \nu_T \not\ll \mu, \quad \nu_T \not\ll \mu, \quad \mu \not\ll \nu_T \\ \sum_k \delta_k = \infty & \nu_T \perp \mu. \end{array} \tag{111}$$

In particular, $\theta_k = 1/k$ gives $T - I \in \mathcal{S}_2$ but $\nu_T \perp \mu$. With $\theta_k = 1/k^2$, the laws instead have positive affinity and neither is absolutely continuous with respect to the other.

For comparison, rescale to variance one a density proportional to $(1 - x^2)^r \mathbf{1}_{\{|x| < 1\}}$, $r > 1$. Its zero-extended square root satisfies the Sobolev hypotheses. Let f be this density, $\rho(dx) = f(x) dx$, $\mu_2 = \rho^{\otimes 2}$, and $g = \sqrt{f}$. Writing $J = 4\|g'\|_2^2 > 1$, the skew-generator formula gives

$$1 - A(\mu_2, (O_\theta)_\# \mu_2) = \frac{J-1}{4} \theta^2 + o(\theta^2). \quad (112)$$

Indeed, for the planar skew generator $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, one has $\|A\|_F^2 = 2$ and $h^2 = \theta^2 \langle A, \mathcal{I}A \rangle_F / 4 + o(\theta^2)$. The only rotations preserving the square support are its signed permutation symmetries. Every rotated square shares a neighborhood of the origin with the original square, on which both densities are positive; hence $A(\mu_2, (O_\theta)_\# \mu_2) > 0$ for every angle. Affinity is continuous and strictly below one away from them, so the three cases in (111) now use $\sum_k \delta_k^2$ in place of $\sum_k \delta_k$. Equivalence still requires every $\delta_k = 0$. Thus the quadratic metric supplied by the Fourier/Fisher method controls accumulation of Hellinger loss, while the zero-set row condition separately controls absolute continuity.

5.4 Complete linear classification for bounded and half-line laws

Write $\rho = \text{Law}(X)$, $\rho^- = \text{Law}(-X)$, and $\mu = \rho^{\otimes \mathbb{N}}$. Throughout this subsection X is centered with variance one. The operator action is the measurable row-series realization of lemma 3.1.

For bounded and one-sided supports, conditional expectations turn the support boundary into an ℓ^1 -type row constraint, and the inverse relations force a monomial operator. No density or Fisher information is needed.

Theorem 5.3 (Linear equivalence for bounded and half-line coordinate laws). *Let $T \in \text{GL}(H)$.*

1. *Suppose $a = \text{ess inf } X < 0 < b = \text{ess sup } X$ are finite. Then $\nu_T \sim \mu$ forces T to be a signed permutation. The complete equivalence group is as follows:*

<i>relation between ρ^- and ρ</i>	<i>allowed signed permutations</i>
$\rho^- = \rho$	<i>all signs</i>
$\rho^- \sim \rho$, $\rho^- \neq \rho$	<i>only finitely many negative signs</i>
$\rho^- \not\sim \rho$	<i>no negative signs.</i>

In particular, $a + b \neq 0$ permits only ordinary permutations. The support need not fill an interval, and a density is not required.

2. *Suppose $\text{ess inf } X = a > -\infty$ and $\text{ess sup } X = +\infty$. Then*

$$\nu_T \sim \mu \iff T \text{ is an ordinary coordinate permutation.} \quad (113)$$

The same conclusion holds for a law bounded above and unbounded below.

In the second case every equivalence is exact invariance. Failure of any of these equivalence criteria does not, by itself, imply singularity.

Proof. Assume $\nu_T \sim \mu$, and set $S = T^{-1}$. The measurable inverse action also gives $\nu_S \sim \mu$. For every row v of T or S , the series $Z = \sum_j v_j X_j$ converges in L^2 , and

$$\mathbb{E}[Z \mid X_1, \dots, X_m] = \sum_{j \leq m} v_j X_j. \quad (114)$$

In the bounded case $Z \in [a, b]$ almost surely, so its conditional expectation does too. The essential width of the finite sum in (114) is $(b - a) \sum_{j \leq m} |v_j|$; hence every row of T and S has ℓ^1 -norm at most one.

Let $A_{ij} = |T_{ij}|$ and $B_{ij} = |S_{ij}|$. The ℓ^2 row/column Cauchy–Schwarz inequality makes the entries of TS and ST absolutely convergent, so $AB \geq I$ and $BA \geq I$ entrywise. Tonelli and the row bounds give

$$\sum_j (AB)_{ij} = \sum_k A_{ik} \sum_j B_{kj} \leq 1, \quad \sum_j (BA)_{ij} \leq 1.$$

Thus $AB = BA = I$. For each i , choose k with $A_{ik}B_{ki} > 0$. The off-diagonal entries of AB force row k of B to vanish outside column i , and those of BA force row i of A to vanish outside column k . This gives a bijective monomial pattern. Its weights and reciprocals are at most one by the row bounds, so every weight is one; hence T is a signed permutation.

For a negative row sign, scalar equivalence requires $\rho^- \sim \rho$, and therefore $a = -b$. Each negative sign contributes $A(\rho, \rho^-)$ to the product affinity. This number equals one exactly when $\rho^- = \rho$; otherwise Kakutani’s theorem [17] permits only finitely many negative signs. This proves the three alternatives.

In the lower-bounded, upper-unbounded case, centering and nondegeneracy give $a < 0$. Every finite sum in (114) is bounded below by a , so no coefficient can be negative. Its essential infimum is $a \sum_{j \leq m} v_j$, whence

$$a \sum_{j \leq m} v_j \geq a, \quad \text{hence} \quad \sum_j v_j \leq 1.$$

Thus T and S are nonnegative with row sums at most one, and the preceding inverse-matrix argument makes T an ordinary permutation. Such permutations preserve μ ; reflection handles the upper-bounded case. $\square$

For $Y \sim \text{Gamma}(r_\Gamma, \theta_\Gamma)$, where $r_\Gamma > 0$ is the shape and $\theta_\Gamma > 0$ is the scale, put

$$X = \frac{Y - r_\Gamma \theta_\Gamma}{\sqrt{r_\Gamma} \theta_\Gamma}.$$

Its essential endpoints are $-\sqrt{r_\Gamma}$ and $+\infty$. Thus (113) gives the complete homogeneous linear equivalence classification for every $r_\Gamma, \theta_\Gamma > 0$, including the shape-two Fourier example in Section 2.4: only ordinary coordinate permutations are allowed. Compact support permits signs only to the extent allowed by the scalar pair ρ, ρ^- . For compactly supported densities positive almost everywhere inside $(-a, a)$, reflection equivalence is automatic, so an asymmetric such density permits finitely many sign changes although they do not preserve the measure exactly. An asymmetric Beta law centered and rescaled to variance one has unequal endpoint magnitudes, so it permits only permutations. These conclusions come from support geometry rather than from the Fourier/Fisher estimates.

5.5 Why the zero-set row condition is indispensable

Even a rank-one perturbation separates positive Hellinger overlap from two-sided absolute continuity.

Let f be the rescaling to variance one of $C(1 - x^2)^2 \mathbf{1}_{\{|x| < 1\}}$, and write its support as $[-a, a]$. Set $\rho(dx) = f(x) dx$. Its zero-extended square root belongs to $W^{1,2}(\mathbb{R})$, and $x(\sqrt{f})' \in L^2(\mathbb{R})$.

For $0 < c < 1$, set

$$T = c \oplus I, \quad f_c(y) = c^{-1} f(y/c), \quad \mu = \rho^{\otimes \mathbb{N}}.$$

Then $T - I$ has rank one and

$$\nu_T = (f_c(y) dy) \otimes \rho^{\otimes \{2,3,\dots\}} \ll \mu, \quad \mu \not\ll \nu_T.$$

Indeed, f_c is positive on $(-ca, ca)$ and zero on the rest of $(-a, a)$. Nevertheless,

$$A(\mu, \nu_T) = \int \sqrt{f(y)f_c(y)} dy > 0, \quad \sup_n E_n(T) < \infty.$$

Thus neither Hilbert–Schmidt proximity nor bounded projected Hellinger energy implies equivalence without the inverse zero-set row condition. The failure occurs with finite location and scale Fisher information.

5.6 Exact block criteria without equivalence of the factors

The boundary examples use a block-product criterion that separates local absolute continuity from accumulated Hellinger loss.

Theorem 5.4 (Block products without equivalence of the factors). *Let $\mu = \bigotimes_{k \geq 1} \lambda_k$ and $\nu = \bigotimes_{k \geq 1} \eta_k$ be probability products over finite-dimensional coordinate blocks. Set*

$$r_k = A(\lambda_k, \eta_k), \quad L = \sum_k -\log r_k \in [0, \infty], \quad -\log 0 = +\infty.$$

Then

$$A(\mu, \nu) = \prod_k r_k, \quad \mu \perp \nu \iff L = \infty. \quad (115)$$

If $L < \infty$, then

$$\begin{aligned} \nu \ll \mu &\iff \eta_k \ll \lambda_k \text{ for every } k, \\ \mu \ll \nu &\iff \lambda_k \ll \eta_k \text{ for every } k. \end{aligned} \quad (116)$$

Thus positive affinity allows equivalence, either direction of strict absolute continuity, or neither direction. For a linear block $A_k \in \text{GL}(d_k)$ and $\lambda_k = \rho^{\otimes d_k}$, where $\rho(dx) = f(x) dx$, the comparison density is

$$|\det A_k|^{-1} \prod_{i=1}^{d_k} f((A_k^{-1}y)_i).$$

The local absolute-continuity conditions concern its actual zero set, not just the closure of its support.

Proof. Affinity tensorizes on finite block prefixes, and projected affinities decrease to global affinity by the conditional-expectation argument in [proposition 5.7](#). This proves (115).

The local absolute-continuity conditions are necessary by projection. If $\eta_k \ll \lambda_k$ for every k and $\prod_k r_k > 0$, set $\ell_n = \prod_{k \leq n} d\eta_k/d\lambda_k$. Independence gives

$$\|\sqrt{\ell_m} - \sqrt{\ell_n}\|_{L^2(\mu)}^2 = 2 \left(1 - \prod_{n < k \leq m} r_k \right) \longrightarrow 0.$$

Their L^2 limit has norm one, and its square defines a probability measure which cylinder tests identify with ν ; hence $\nu \ll \mu$. Interchanging μ and ν proves the reverse assertion. The local condition cannot be omitted [\[17, 32\]](#). $\square$

Corollary 5.5 (Masses of the absolutely continuous parts). *Take $m_k = (\lambda_k + \eta_k)/2$, with densities p_k, q_k , and put*

$$a_k = \lambda_k(\{q_k > 0\}), \quad b_k = \eta_k(\{p_k > 0\}).$$

If $L < \infty$, denote the absolutely continuous parts of ν relative to μ , and of μ relative to ν , by ν_a and μ_a , respectively. Then

$$\nu_a(\mathbb{R}^{\mathbb{N}}) = \prod_k b_k, \quad \mu_a(\mathbb{R}^{\mathbb{N}}) = \prod_k a_k. \quad (117)$$

Proof. Since $r_k^2 \leq a_k b_k$ and $\prod_k r_k > 0$, both displayed products are positive. On $C = \prod_k \{p_k q_k > 0\}$, the conditional product laws have equivalent factors and affinities $r_k/\sqrt{a_k b_k}$ with positive product, so [theorem 5.4](#) makes them equivalent. The restrictions to C are therefore precisely the absolutely continuous parts, with masses $\prod_k b_k$ and $\prod_k a_k$, respectively. $\square$

Gamma scale changes and the effect of centering. Fix $r_\Gamma > 0$. For the uncentered shape- r_Γ , scale-one Gamma density, positive scaling by c preserves the scalar measure class and gives

$$\int_0^\infty \sqrt{f(y)c^{-1}f(y/c)} dy = \left(\frac{2\sqrt{c}}{1+c} \right)^{r_\Gamma} = \cosh\left(\frac{\log c}{2}\right)^{-r_\Gamma}. \quad (118)$$

Integration of $y^{r_\Gamma-1}e^{-(1+c^{-1})y/2}$ proves the formula. Since $\log \cosh(u/2) \sim u^2/8$ at zero, it follows that

$$\bigotimes_k \text{Law}(c_k Y) \sim \bigotimes_k \text{Law}(Y) \iff \sum_k (\log c_k)^2 < \infty, \quad c_k > 0. \quad (119)$$

The condition is equivalently $\sum_k (c_k - 1)^2 < \infty$. This does not contradict [theorem 5.3](#): for $X = (Y - r_\Gamma)/\sqrt{r_\Gamma}$, the map $Y \mapsto cY$ becomes

$$X \mapsto cX + \sqrt{r_\Gamma}(c - 1),$$

an affine map fixing the lower endpoint, not the homogeneous linear map $X \mapsto cX$.

Corollary 5.6 (Coordinatewise affine equivalence for Gamma products). *Let μ be the product of the shape- r_Γ Gamma law centered with variance one. For $T \in \text{GL}(H)$ and $b \in \ell^2$, the law of $\Phi_T(x) + b$ under μ is equivalent to μ if and only if there are a coordinate permutation π and numbers $c_i > 0$ such that*

$$(Tx)_i = c_i x_{\pi(i)}, \quad b_i = \sqrt{r_\Gamma}(c_i - 1), \quad \sum_i (\log c_i)^2 < \infty.$$

Proof. The inverse affine map has linear part T^{-1} and shift $-T^{-1}b \in \ell^2$. For the row-series realizations, Cauchy–Schwarz gives $\Phi_S(z + b) = \Phi_S(z) + Sb$ wherever the selected row limits exist, for every $b \in \ell^2$. The affine orbit satisfies the covariance bound required by [lemma 3.1](#); hence that lemma and the translation identity show that the inverse affine map also preserves the measure class. Conditioning finite row sums, as in [theorem 5.3](#), shows that both T and T^{-1} have nonnegative entries: a negative coefficient would create an unbounded lower tail that no fixed shift can remove. The inverse identities then make T monomial with positive weights c_i . Equality of the scalar lower endpoints forces $-\sqrt{r_\Gamma}c_i + b_i = -\sqrt{r_\Gamma}$. Now (119) proves necessity and sufficiency. $\square$

5.7 An exact finite-marginal equivalence criterion

The following finite-marginal criterion isolates the measure-theoretic content without Sobolev assumptions or a dichotomy; under (105), theorem 5.2 specializes it to bounded invertible linear mixing.

Proposition 5.7 (Finite-marginal tails characterize equivalence). *Let μ, ν be any probability measures on $\mathbb{R}^{\mathbb{N}}$, and let μ_n, ν_n be their first n marginals. Choose densities p_n, q_n with respect to any common dominating measure m_n . For $M > 1$, define*

$$\text{Tail}_n(M) = \int_{\{q_n > Mp_n\}} q_n dm_n + \int_{\{p_n > Mq_n\}} p_n dm_n.$$

Then

$$\mu \sim \nu \iff \lim_{M \rightarrow \infty} \sup_n \text{Tail}_n(M) = 0. \quad (120)$$

Equivalently, every finite pair is equivalent and its forward and reverse likelihood-ratio martingales are uniformly integrable in their respective reference measures. Without any finite-marginal equivalence assumption,

$$A(\mu_n, \nu_n) \downarrow A(\mu, \nu), \quad \sup_n [-2 \log A(\mu_n, \nu_n)] < \infty \iff \mu \not\sim \nu. \quad (121)$$

Proof. If the limit in (120) is zero, a set where $p_n = 0 < q_n$ must have zero ν_n -mass for every n ; the second integral gives the reverse, so $\mu_n \sim \nu_n$. With $\ell_n(x) = q_n(\pi_n x)/p_n(\pi_n x)$, the two summands in $\text{Tail}_n(M)$ are

$$\mathbb{E}_\mu[\ell_n \mathbf{1}_{\{\ell_n > M\}}], \quad \mathbb{E}_\nu[\ell_n^{-1} \mathbf{1}_{\{\ell_n^{-1} > M\}}].$$

Thus the displayed limit is exactly uniform integrability of the two mean-one likelihood-ratio martingales. Their L^1 limits have mass one, and cylinder tests identify the corresponding absolutely continuous measures, proving mutual absolute continuity. Conversely, under equivalence the martingales are conditional expectations of the full likelihood ratio and its reciprocal, and are therefore uniformly integrable.

For the affinity assertion put $m = (\mu + \nu)/2$, $u = d\mu/dm$, and $v = d\nu/dm$. For $\mathcal{F}_n = \sigma(x_1, \dots, x_n)$, set $u_n = \mathbb{E}_m[u \mid \mathcal{F}_n]$ and $v_n = \mathbb{E}_m[v \mid \mathcal{F}_n]$. Since $u, v \leq 2$, martingale convergence and bounded convergence give

$$A(\mu_n, \nu_n) = \int \sqrt{u_n v_n} dm \longrightarrow \int \sqrt{uv} dm.$$

Conditional Cauchy–Schwarz gives monotonicity, and global affinity vanishes exactly for mutually singular measures. This proves (121). $\square$

For bounded invertible linear mixing, finite output marginals still have Lebesgue densities, though their zero sets may differ from those of the product density. Thus (120) remains exact; theorems 5.2 and 5.3 turn it into explicit operator classifications in the regular zero-set and support-boundary regimes.

5.8 What the Finite-Variance Hypotheses Control

It remains to isolate the roles of the analytic hypotheses within the positive-density regime.

The a.e.-positive-density theorem includes the centered, variance-one Laplace density despite its cusp, as well as the normalized density proportional to e^{-x^4} ; symmetry is never used.

For examples allowing both asymmetry and heavy tails, take a density proportional to

$$(1 + y^2)^{-(\tau+1)/2} e^{\vartheta \arctan y}, \quad \tau > 2, \quad \vartheta \in \mathbb{R}, \quad (122)$$

and center and rescale to variance one. Its location score before centering and rescaling is

$$\frac{(\tau + 1)y - \vartheta}{1 + y^2}.$$

The score and its product with y are bounded, so centering and rescaling give an everywhere-positive example satisfying (1). For $2 < \tau \leq 4$ its fourth moment is infinite, and for $\vartheta \neq 0$ its two tails have unequal leading constants. Thus the theorem assumes neither evenness nor a fourth moment; $\vartheta = 0$ recovers a rescaled Student t family.

The centered Gamma example has a half-line zero set: its rotational Fourier defect detects local rotations, while the support theorem leaves only coordinate permutations for homogeneous linear equivalence. [Corollary 5.6](#) treats the distinct affine problem, in which support-preserving scales can remain equivalent.

The hypotheses enter at different points. Support controls null-set admissibility. The second moment makes every ℓ^2 row a convergent random series, supplies uniform integrability of squared linear forms, and enables the Gaussian-summand extraction in [lemma 3.8](#). Hence the first-moment ODE result has strictly broader moment scope than [theorem 1.1](#).

Location and scale Fisher information control the Hellinger paths and their dimension-free generator bounds. The test functions H_A require no higher moments: their role is to pair those bounds with every matrix tangent. Tail symmetry then avoids a separate mixed-law variance estimate. This division of labor explains why the orthogonal theorem drops scale Fisher information, while the full bounded invertible theorem retains it.

References

- [1] Shun-ichi Amari and Hiroshi Nagaoka. *Methods of Information Geometry*, volume 191 of *Translations of Mathematical Monographs*. American Mathematical Society and Oxford University Press, Providence, RI, and Oxford, 2000. doi: 10.1090/mmono/191.
- [2] L. M. Arutyunyan and E. D. Kosov. Spaces of quasi-invariance of product measures. *Functional Analysis and Its Applications*, 49(2):142–144, 2015. doi: 10.1007/s10688-015-0096-x.
- [3] Denis Bell. A quasi-invariance theorem for measures on Banach spaces. *Transactions of the American Mathematical Society*, 290(2):851–855, 1985. doi: 10.1090/S0002-9947-1985-0792833-3.
- [4] Sergey G. Bobkov, Gennadiy P. Chistyakov, and Friedrich Götze. Fisher information and convergence to stable laws. *Bernoulli*, 20(3):1620–1646, 2014. doi: 10.3150/13-BEJ535.
- [5] Vladimir I. Bogachev. Differentiable measures and the Malliavin calculus. *Journal of Mathematical Sciences*, 87(4):3577–3731, 1997. doi: 10.1007/BF02355450.

- [6] Vladimir I. Bogachev. *Gaussian Measures*, volume 62 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 1998. doi: 10.1090/surv/062.
- [7] Robert H. Cameron and William T. Martin. Transformations of Wiener integrals under translations. *Annals of Mathematics*, 45(2):386–396, 1944. doi: 10.2307/1969276.
- [8] Pierre Comon. Independent component analysis, a new concept? *Signal Processing*, 36(3):287–314, 1994. doi: 10.1016/0165-1684(94)90029-9.
- [9] Georges Darmois. Analyse générale des liaisons stochastiques: étude particulière de l’analyse factorielle linéaire. *Revue de l’Institut International de Statistique*, 21(1/2):2–8, 1953. doi: 10.2307/1401511.
- [10] Jacob Feldman. Equivalence and perpendicularity of Gaussian processes. *Pacific Journal of Mathematics*, 8(4):699–708, 1958. doi: 10.2140/pjm.1958.8.699. URL <https://projecteuclid.org/euclid.pjm/1103039696>.
- [11] Jacob Feldman. Examples of non-Gaussian quasi-invariant distributions in Hilbert space. *Transactions of the American Mathematical Society*, 99(2):342–349, 1961. doi: 10.2307/1993405.
- [12] Jaroslav Hájek. On a property of normal distributions of any stochastic process. *Czechoslovak Mathematical Journal*, 8(4):610–618, 1958. doi: 10.21136/CMJ.1958.100333. In Russian; English title translation.
- [13] Aoi Honda, Yoshiaki Okazaki, and Hiroshi Sato. Approximations and the linearity of the Shepp space. *Kyushu Journal of Mathematics*, 69(1):173–194, 2015. doi: 10.2206/kyushujm.69.173.
- [14] Akihito Hora. Quasi-invariant measures on commutative Lie groups of infinite product type. *Mathematische Zeitschrift*, 206(2):169–192, 1991. doi: 10.1007/BF02571334.
- [15] Mark Kac. On a characterization of the normal distribution. *American Journal of Mathematics*, 61(3):726–728, 1939. doi: 10.2307/2371328.
- [16] Abram M. Kagan, Yuri V. Linnik, and Callyampudi Radhakrishna Rao. *Characterization Problems in Mathematical Statistics*. Wiley Series in Probability and Mathematical Statistics. John Wiley & Sons, New York, 1973. ISBN 0471454214.
- [17] Shizuo Kakutani. On equivalence of infinite product measures. *Annals of Mathematics*, 49(1):214–224, 1948. doi: 10.2307/1969123.
- [18] Kazuhiro Kitada and Hiroshi Sato. On the absolute continuity of infinite product measure and its convolution. *Probability Theory and Related Fields*, 81(4):609–627, 1989. doi: 10.1007/BF00367307.
- [19] Lucien Le Cam. *Asymptotic Methods in Statistical Decision Theory*. Springer Series in Statistics. Springer-Verlag, New York, 1986. doi: 10.1007/978-1-4612-4946-7.
- [20] Han Cheng Lie and T. J. Sullivan. Quasi-invariance of countable products of Cauchy measures under non-unitary dilations. *Electronic Communications in Probability*, 23:1–6, 2018. doi: 10.1214/18-ECP113. Paper no. 8.

- [21] Friedrich Liese and Igor Vajda. On divergences and informations in statistics and information theory. *IEEE Transactions on Information Theory*, 52(10):4394–4412, 2006. doi: 10.1109/TIT.2006.881731.
- [22] V. Mandrekar. Lebesgue decomposition of probability measures: Applications to equivalence, singularity and kriging. *Expositiones Mathematicae*, 36(3–4):351–361, 2018. doi: 10.1016/j.exmath.2018.06.006.
- [23] Mauro S. de F. Marques and Stamatis Cambanis. Dichotomies for certain product measures and stable processes. *Probability and Mathematical Statistics*, 12(2):271–289, 1991. URL <https://www.math.uni.wroc.pl/~pms/publications.php?nr=12.2>.
- [24] Mauro S. de F. Marques and Luiz A. B. San Martin. Quasi-invariance of product measures under Lie group perturbations: Fisher information and L^2 -differentiability. *Annals of Probability*, 20(3):1420–1435, 1992. doi: 10.1214/aop/1176989697.
- [25] John P. Nolan. *Univariate Stable Distributions: Models for Heavy Tailed Data*. Springer Series in Operations Research and Financial Engineering. Springer, Cham, 2020. doi: 10.1007/978-3-030-52915-4.
- [26] Kazuki Okamura. An equivalence criterion for infinite products of Cauchy measures. *Statistics & Probability Letters*, 163:108797, 2020. doi: 10.1016/j.spl.2020.108797.
- [27] Yoshiaki Okazaki. Equivalent-singular dichotomy for quasi-invariant ergodic measures. *Annales de l’Institut Henri Poincaré, Probabilités et Statistiques*, 21(4):393–400, 1985. URL https://www.numdam.org/item/AIHPB_1985__21_4_393_0/.
- [28] Roald Ramer. On nonlinear transformations of Gaussian measures. *Journal of Functional Analysis*, 15(2):166–187, 1974. doi: 10.1016/0022-1236(74)90017-2.
- [29] Gaku Sadasue. On quasi-invariance of infinite product measures. *Journal of Theoretical Probability*, 21(3):571–585, 2008. doi: 10.1007/s10959-008-0171-9.
- [30] Gennady Samorodnitsky and Murad S. Taqqu. *Stable Non-Gaussian Random Processes: Stochastic Models with Infinite Variance*. Chapman & Hall, New York, 1994. ISBN 9780412051715.
- [31] Ken-iti Sato. *Lévy Processes and Infinitely Divisible Distributions*, volume 68 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 1999. ISBN 9780521553025.
- [32] Lawrence A. Shepp. Distinguishing a sequence of random variables from a translate of itself. *Annals of Mathematical Statistics*, 36(4):1107–1112, 1965. doi: 10.1214/aoms/1177699985. URL <https://www.jstor.org/stable/2238114>.
- [33] Lawrence A. Shepp. Radon–Nikodym derivatives of Gaussian measures. *Annals of Mathematical Statistics*, 37(2):321–354, 1966. doi: 10.1214/aoms/1177699516.
- [34] Hiroaki Shimomura. Linear transformation of quasi-invariant measures. *Publications of the Research Institute for Mathematical Sciences*, 12(3):777–800, 1977. doi: 10.2977/PRIMS/1195190378.

- [35] Barry Simon. *Trace Ideals and Their Applications*, volume 120 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, second edition, 2005. doi: 10.1090/surv/120.
- [36] V. P. Skitovich. Linear forms of independent random variables and the normal distribution law. *Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya*, 18(2):185–200, 1954. URL <https://www.mathnet.ru/eng/im3497>. In Russian; English title translation.
- [37] J. Michael Steele. Fisher information and detection of a Euclidean perturbation of an independent stationary process. *Annals of Probability*, 14(1):326–335, 1986. doi: 10.1214/aop/1176992631.
- [38] Yasuji Takahashi. Quasi-invariant measures on linear topological spaces. *Hokkaido Mathematical Journal*, 4(1):59–81, 1975. doi: 10.14492/hokmj/1381758785.
- [39] Aad W. van der Vaart. *Asymptotic Statistics*, volume 3 of *Cambridge Series in Statistical and Probabilistic Mathematics*. Cambridge University Press, Cambridge, 1998. doi: 10.1017/CBO9780511802256.
- [40] Vladimir M. Zolotarev. *One-Dimensional Stable Distributions*, volume 65 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1986. doi: 10.1090/mmono/065.